

Inflation-Indexed Debt and the Risks of Fiscal Dominance*

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Abstract

The origins of the post-2020 inflation are the subject of much debate. One argument has its roots in an unfunded expansion of debt-driven government spending, in what has been labelled *fiscal dominance* (Leeper, 1991) or a *fiscally-led policy mix* (Bianchi et al., 2023). We show that the risks of such fiscal dominance depend on the degree to which government debt is indexed to inflation. Inflation-indexation has a nonlinear effect on the existence of saddlepath equilibria, and can amplify the inflationary effects of deficit shocks when policy is fiscally led unless monetary and fiscal policy enact coordinated policies that limit, or even invert, the windfall gains borne by households holding indexed debt. Empirical evidence links inflation-indexed debt to low central bank independence, a high probability of suspending fiscal rules, and a larger reaction of inflation to fiscal shocks.

Keywords: Fiscal-Monetary Interactions, Fiscal Dominance, Sovereign Debt.

JEL Codes: E30, E63.

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1 Introduction

Recent high inflation has renewed interest in the interactions between fiscal and monetary policy. The inflation experienced in the U.S. in 2021-23 coincided with a large debt-financed fiscal expansion that, if not accompanied by corresponding promises to repay the debt in equivalent real terms in the future, must lead to a depreciation in the real value of government debt through an inflation-driven debt devaluation process. Current stubbornly persistent inflation, likewise, occurs alongside continued deficits without a conceivable plan to limit their magnitude in the medium-term. The possibility of fiscal policy acting persistently in this way has contributed to concerns about the sustainability of government finances, especially given record absolute debt levels and debt-to-GDP ratios only previously seen in the aftermath of wars.¹

This paper shows that, conditional on a lack of policy coordination accounting for the presence of indexed debt, the risks associated with fiscal dominance are heightened when the face value of at least some of a government's outstanding debt is automatically uprated by the gross rate of inflation, as with Treasury Inflation-Protected Securities (TIPS) in the U.S. and Index-Linked Gilts in the U.K. We find that inflation-indexed debt has a nonlinear impact on the existence of saddle path-stable equilibria, and that it can induce more volatility in inflation if there are deficit-driven government transfer shocks and policy is fiscally-led. In short, fiscal dominance is often more consequential when a higher proportion of government debt is indexed to inflation.

That inflation-indexed debt affects the risks of fiscal dominance has so far not been recognised. Although a number of theoretical ([Angeletos et al., 2024](#); [Ascari et al., 2023](#); [Bianchi et al., 2023](#); [Cochrane, 2022](#)) and empirical ([Barro and Bianchi, 2026](#); [Hazell and Hobler, 2024](#); [Hilscher et al., 2022](#)) papers explore the recent inflationary episode using models that emphasise fiscal-monetary interactions and the extent to which debt-driven fiscal expansions can fuel inflation, they all abstract from inflation-indexed debt and so overlook the role it plays in determining the risks and consequences of fiscal dominance and a fiscally-led policy mix.²

We begin with a simple Fisherian environment where households save using government debt that is partially indexed to inflation. Following the analytical approach in [Leeper \(1991\)](#), we show that inflation-indexed debt restricts the admissible space of monetary policies that supports a fiscally-led policy mix. This occurs because inflation-indexed debt imposes an additional constraint on fiscal policy, namely that it must fund any uprating of face values required in response to inflation. We then progress to a non-Ricardian general equilibrium model in the spirit of [Blanchard \(1985\)](#) and [Angeletos et al. \(2024\)](#), where we find that inflation-indexed debt tends to amplify the reaction of inflation to debt-financed transfer shocks. The relationship is nonlinear since it flattens out and even inverts at high levels of inflation-indexed debt.

¹The increased burden of servicing debt is one such area of concern. For example, interest payments on federal debt in the U.S. are expected to reach 7% of GDP by 2050.

²Complementary explanations for the 2021-2023 inflationary episode include broad supply shocks and bounded rationality ([Benigno, 2022](#); [Beaudry et al., 2025](#)) or shocks to commodity prices with sticky labor ([Bernanke and Blanchard, 2023](#)).

It is challenging to use empirical data to establish whether the policy mix in a particular period is monetary or fiscally-led (Cochrane, 2011; Neumeyer and Nicolini, 2025). In the appendix, we offer three distinct pieces of evidence to suggest that inflation-indexed debt plays a role in inflation dynamics through the risks of fiscal dominance. First, using a long sample of global data, we see that the share of inflation-indexed debt correlates positively with inflation and negatively with the Central Bank Independence Index of Romelli (2024). Second, we use cross-country data from Davoodi et al. (2022) to show that fiscal rules are more likely to be suspended in countries that have a high proportion of government debt indexed to inflation. Third, we combine the narratively-identified U.S. tax shocks of Mierzwa (2024) with the classification of policy in Chen et al. (2022) to demonstrate that the inflationary effect of fiscal shocks is only meaningfully amplified by indexed debt when policy is fiscally-led.

The paper is organised as follows. Section 2 introduces inflation-indexed debt in a simple Fisherian model, deriving determinacy conditions and characterising the behaviour of inflation when the policy mix is fiscally led. Section 3 deepens the analysis of inflation-indexed debt in a general equilibrium setting, while section 4 highlights the scope of monetary and fiscal policymakers to un-do the possibly detrimental effects of indexed debt. Section 5 concludes, while some empirical evidence in favour of our mechanisms can be found in appendix C.

Literature Review

This paper contributes to the burgeoning literature on fiscal-monetary interactions, pioneered in Sargent and Wallace (1981) and formalized through Leeper (1991). Leeper and Leith (2016) and Cochrane (2023) summarize the state of the literature. Empirical and theoretical support for the possibility of fiscally-driven inflation has recently been developed in Barro and Bianchi (2026) and Cloyne et al. (2023), mirroring the interest in possible fiscal drivers in the recent inflationary episode. In particular, Bianchi et al. (2023) and Smets and Wouters (2024) extensively analyse the importance of the interactions between fiscal and monetary policy for fiscally-driven inflation in canonical general equilibrium models. Additional evidence utilizing novel model dimensions, such as monetary unions, interest rate boundaries, and game-theoretic interactions between fiscal and monetary policymakers are provided by Reichlin et al. (2023); Miao and Su (2024), and Corsetti and Maćkowiak (2024).

Closest related to us in terms of our modelling framework are Angeletos et al. (2024) and Rachel and Ravn (2025), who themselves work with a model building on the seminal contribution of Blanchard (1985), facilitating a tractable way of introducing non-Ricardian fiscal policies using simple transfer shocks, which do not directly alter the resource constraint present in the economy.³ Angeletos et al. (2024) and Rachel and Ravn (2025) prove the conditions for existence and uniqueness of dynamic competitive equilibria in a NK-OLG model, delivering boundary conditions under

³Working with government spending shocks, for instance, would alter the resource constraint in the economy. Without further assumptions on the household utility borne by such government spending, however, any such government spending would be ‘wasteful’.

which equilibria can be characterized. [Rachel and Ravn \(2025\)](#) deliver two particularly insightful results related to our work; first, indeterminacy becomes a smaller issue once households are non-Ricardian, i.e., adherence to the Taylor principle is not strictly necessary to deliver unique saddle-path stable equilibria even when fiscal policy is conventionally ‘passive’; and second, the classic distinction between ‘active’ and ‘passive’ policies should not be taken literally once households are non-Ricardian. Relative to this body of work, our additional consideration of inflation-indexed debt innovates on a number of dimensions. First, inflation-indexed debt overcomes the singularity result of [Angeletos et al. \(2024\)](#) that RANK-FTPL and HANK models converge to the same limit point; second, inflation-indexed debt generally increases the response of inflation in response to transfer shocks; and finally, inflation-indexed debt alters one of the conclusions of [Rachel and Ravn \(2025\)](#), namely that the explosive region becomes more problematic under non-Ricardianness, providing an additional anchor favouring determinate equilibria. The latter part of this paper and the evaluation of policy rules accounting for the presence of inflation-indexed debt is also clearly related to [Holden \(2024\)](#), who explores mechanisms under which central banks can target desired nominal outcomes through appropriate policy rule specifications.

[Banerjee et al. \(2022\)](#) evaluates the prevalence of fiscally-driven policy mixes empirically with a focus on a wider set of advanced economies. A narrative example of a recent fiscal shock informing inflation rates is provided by [Hazell and Hobler \(2024\)](#), who focus on the 2021 Georgia Senate election run-off, with similar supplementary evidence on a specific policy shock (the ‘Truss shock’) in the UK being provided by [Kawalec \(2025\)](#).

Albeit indirectly, we also contribute to the literature on inflation-linked government bonds. [Fischer \(1975\)](#) derived household demand for indexed bonds in a multi-asset optimal allocation framework. The special insurance properties of such inflation-linked debt are extensively discussed in [Barr and Campbell \(1997\)](#), [Garcia and van Rixtel \(2007\)](#), [Gürkaynak et al. \(2010\)](#) and [Andreasen et al. \(2021\)](#), and their usefulness for policy has been studied by [Falcetti and Missale \(2002\)](#), [Persson et al. \(2006\)](#), and [Hatcher \(2014\)](#). Notably, [Sims \(2013\)](#) briefly mentions the possible detrimental consequences of indexed debt in frameworks that extensively feature fiscally-driven policy mixes. This paper expands on his remarks using simple, analytical modelling frameworks. [Schmid et al. \(2026\)](#) provide a systematic analysis of inflation-indexed debt as a policy tool, emphasizing its role as an ex-ante commitment device against inflation. With our contribution, we do not invalidate the conclusions of [Schmid et al. \(2026\)](#) about the desire for indexed bonds as plausibly overcoming a certain degree of market incompleteness, but we instead add to their results by qualifying the possibility of risks associated with the issuance of inflation-indexed debt that have henceforth been hidden from the literature. In complementary work ([Kawalec, 2025](#)), we focus on a cross-country comparison of the effects of indexed debt, putting quantitative estimates at the forefront in a richer heterogeneous-agent model. In contrast, this paper focuses on the narrower question of the theoretical properties of indexed debt in macroeconomic general equilibrium models, with supplementary evidence that highlights the potential to explore strategic interactions between monetary and fiscal policymakers more clearly.

2 Example from a Fisherian model

We illustrate how inflation-indexed debt matters in a canonical Fisherian model with a representative household saving in a government bond that is partially indexed to inflation. We do so to analyse how the presence of indexed debt can influence the equilibrium properties in dynamic economies as simply as possible.

A representative household receives a constant stream of goods Y , maximizing its expected present value of utility from consumption, discounted at the time-invariant rate β :

$$\max_{\{c_t, b_t\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

subject to the flow budget constraint

$$P_t c_t + q_t b_t = P_t(Y - T_t) + \Pi_t^\theta b_{t-1}, \quad (1)$$

where b_t is an asset that is a *partially*-indexed government-issued security. The asset is indexed to the gross rate of inflation by θ , in a manner that is reflective of how the face value of indexed debt is formed empirically (Hall and Sargent, 2011). The government-issued asset is bought at the market price q_t . T_t denotes real lump-sum taxes raised by the government.

The household optimality conditions are standard:

$$\begin{aligned} \{c_t\} : \quad & u'(c_t) = \lambda_t P_t \\ \{b_t\} : \quad & \lambda_t q_t = \beta E_t[\lambda_{t+1} \Pi_{t+1}^\theta] \end{aligned}$$

The asset pricing equation is:

$$q_t = \beta E_t \left[\frac{u'(c_{t+1})}{u'(c_t)} \Pi_{t+1}^{\theta-1} \right]. \quad (2)$$

The flow budget constraint of the government is:

$$\Pi_t^\theta b_{t-1} = P_t T_t + q_t b_t, \quad (3)$$

with the bond pricing kernel being given by the household first-order conditions and T_t denoting real surpluses raised by the government, which are equivalent to lump-sum taxes raised. The fiscal policy rule reacts to the debt-to-GDP ratio by adjusting tax rate τ_t :

$$\frac{\tau_t}{\tau} = \left(\frac{s_{t-1}}{s} \right)^\gamma e^{\varphi_t}; \quad s_t \equiv \frac{q_t b_t}{P_t Y}, \quad \tau_t \equiv \frac{T_t}{Y},$$

where $\tau_t \equiv \frac{T_t}{Y}$ are surpluses raised by the government as a fraction of output, and $s_t \equiv \frac{q_t b_t}{P_t Y}$ is

the market value of the government-issued partially-indexed bond relative to GDP. φ_t is a standard AR(1) shock to the present lump-sum tax rate (the exogenous fiscal policy disturbance in the model), and the fiscal policy reaction coefficient is γ .

The monetary policy rule targets the return $R_t = 1/q_t$ on the partially-indexed bond:

$$R_t = \frac{1}{\beta} \Pi_t^\phi. \quad (4)$$

For completeness, we explore the model properties under a standard Taylor Rule in appendix A.2.

Equilibrium

Combining the household budget constraint (1) with the government budget constraint (3), consumption must be equal to the endowment $c_t = Y \forall t$, and the government bond pricing kernel simplifies to $q_t = \beta E_t[\Pi_{t+1}^{\theta-1}]$, reflecting the partial indexation of the government-issued asset and the resulting dependence of its price on expected inflation.

We linearize the equilibrium conditions around a deterministic zero-inflation steady-state ($\Pi = 1$), consistent with household optimality and the fiscal and monetary rules. We denote variables in their log-deviations from steady-state with hats. The detailed derivations can be found in appendix A.1. The system can be expressed up to first order in just two variables: deviations of inflation from steady-state ($\hat{\pi}_t$) and deviations in the market value of debt relative to GDP from steady-state ($\hat{s}_t$):

$$\underbrace{\begin{pmatrix} 1-\theta & 0 \\ 0 & \beta \end{pmatrix}}_{\equiv A_0} E_t \begin{pmatrix} \hat{\pi}_{t+1} \\ \hat{s}_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \phi & 0 \\ 0 & 1-\gamma(1-\beta) \end{pmatrix}}_{\equiv A_1} \begin{pmatrix} \hat{\pi}_t \\ \hat{s}_t \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ -(1-\beta) \end{pmatrix}}_{\equiv C} E_t \varphi_{t+1}. \quad (5)$$

The matrix relevant for the determinacy properties is

$$Z \equiv A_0^{-1} A_1 = \begin{pmatrix} \frac{\phi}{1-\theta} & 0 \\ 0 & \frac{1-\gamma(1-\beta)}{\beta} \end{pmatrix}. \quad (6)$$

Its eigenvalues are $\frac{\phi}{1-\theta}$ and $\frac{1-\gamma(1-\beta)}{\beta}$. A *monetary-led* equilibrium requires $\phi > 1-\theta$ and $\gamma > 1$. A *fiscally-led* equilibrium requires $\phi < 1-\theta$ and $\gamma < 1$. The introduction of inflation-indexed debt hence matters only for the monetary, not fiscal, side of the model.

Figure 1 visualizes the determinacy properties of the model. In a nutshell, the presence of inflation-indexed debt shifts how *monetary* policy impacts the existence of a unique, saddle path-stable equilibrium. Once the share of inflation-indexed debt turns positive (right panel), the space under which a unique monetary-led equilibrium exists increases; that is, central banks can follow "more passive" monetary policies without running into the indeterminate (yellow) space. At the same

time, a fiscal authority wishing to engage in activist fiscal policy without negative repercussions on the stability properties of the system must be aware that this requires an even more 'passive' monetary authority relative to conventional wisdom.

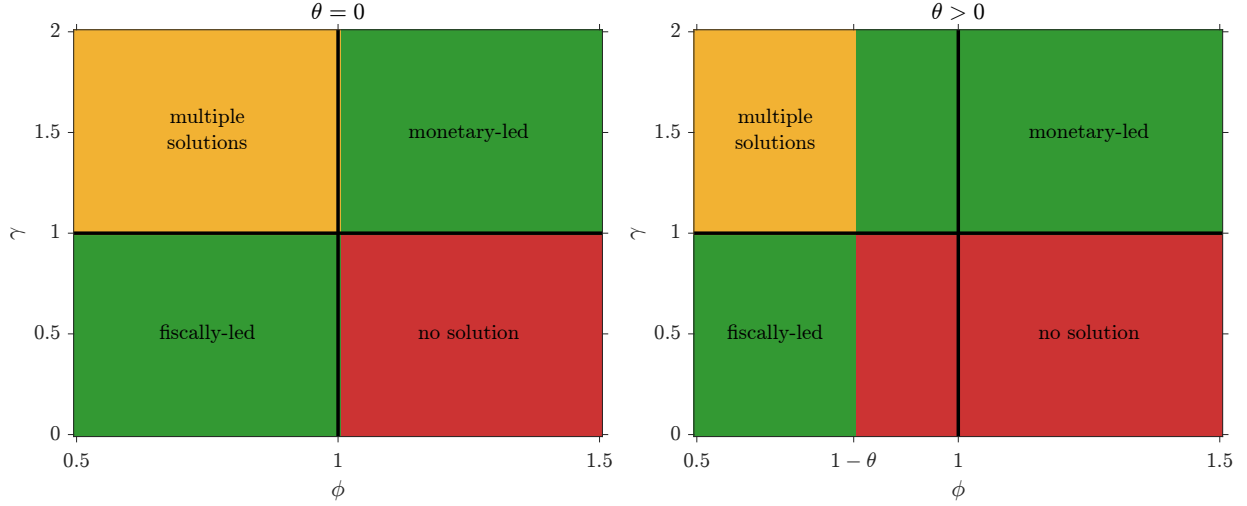


Figure 1: Determinacy properties of the Fisherian model.

The intuition behind this phenomenon is that indexed debt works as an 'automatic stabilizer': as the central bank targets the return rate of the partially-indexed bond, any change it induces on its net return rate has *more* stabilizing effects relative to a canonical Taylor Rule, as a presumed increase in the net policy rate directly decreases the expected future inflationary pressure, thereby stabilizing the system beyond the simple first-order effect of the nominal policy rate adjustment.

To derive debt-inflation dynamics under the fiscally-led policy mix, the block-recursivity of the model means we need to solve the debt equation forward.⁴ The debt equation is:

$$(1 - (1 - \beta)\gamma) \hat{s}_t = (1 - \beta) \mathbb{E}_t \varphi_{t+1} + \mathbb{E}_t \beta \hat{s}_{t+1}.$$

so solving forward implies:

$$\hat{s}_t = \sum_{i=0}^{\infty} \left(\frac{\beta}{1 - (1 - \beta)\gamma} \right)^i \left(\frac{1 - \beta}{1 - (1 - \beta)\gamma} \right) \mathbb{E}_t \varphi_{t+i+1}.$$

When φ_t is an AR(1) process with persistence ρ then $\mathbb{E}_t \varphi_{t+i+1} = \rho^{i+1} \varphi_t$ and:

$$\hat{s}_t = \sum_{i=0}^{\infty} \left(\frac{\beta \rho}{1 - (1 - \beta)\gamma} \right)^i \left(\frac{(1 - \beta)\rho}{1 - (1 - \beta)\gamma} \right) \varphi_t = \frac{(1 - \beta)\rho}{1 - (1 - \beta)\gamma - \beta \rho} \varphi_t,$$

⁴The solution for the monetary-led policy mix trivially features $\pi_t = 0 \forall t$, as otherwise inflation would be explosive. The value of debt under a monetary-led policy mix evolves as $\hat{s}_t = \frac{1 - (1 - \beta)\gamma}{\beta} \hat{s}_{t-1} - \frac{1 - \beta}{\beta} \varphi_t$.

which is independent of θ .

We solve for inflation dynamics in equilibrium by taking the linearized budget constraint and substituting in for the monetary policy rule to obtain:

$$\hat{\pi}_t = \frac{\phi}{1-\theta} \hat{\pi}_{t-1} - \frac{\beta}{1-\theta} \hat{s}_t + \frac{1-(1-\beta)\gamma}{1-\theta} \hat{s}_{t-1} - \frac{1-\beta}{1-\theta} \varphi_t. \quad (7)$$

For an economy initially in steady state, $\hat{s}_{-1} = \hat{\pi}_{-1} = 0$, and a normalized deficit shock $\varphi_0 = -1$, the period-0 government budget constraint pins down the initial inflation jump. Defining $\kappa \equiv \frac{(1-\beta)\rho}{1-(1-\beta)\gamma-\beta\rho}$, so that $\hat{s}_t = \kappa\varphi_t$, we obtain:

$$\hat{\pi}_0 = \frac{\beta\kappa + 1 - \beta}{1 - \theta}.$$

For $t \geq 1$, the debt solution implies that the remaining terms in the inflation equation cancel, such that inflation evolves according to:

$$\hat{\pi}_t = \frac{\phi}{1-\theta} \hat{\pi}_{t-1} = \left(\frac{\phi}{1-\theta} \right)^t \hat{\pi}_0.$$

The tax rate follows $\hat{\tau}_t = \gamma\hat{s}_{t-1} + \varphi_t$, implying $\hat{\tau}_0 = -1$ for the normalized deficit shock. Inflation-indexed debt here has the expected effect of scaling *surprise* inflation: When $E_{t-1}\hat{\pi}_t \neq \hat{\pi}_t$, the fact that the equilibrium price for indexed debt had been *too low* in period $t-1$ induces additional surprise inflation in period t .

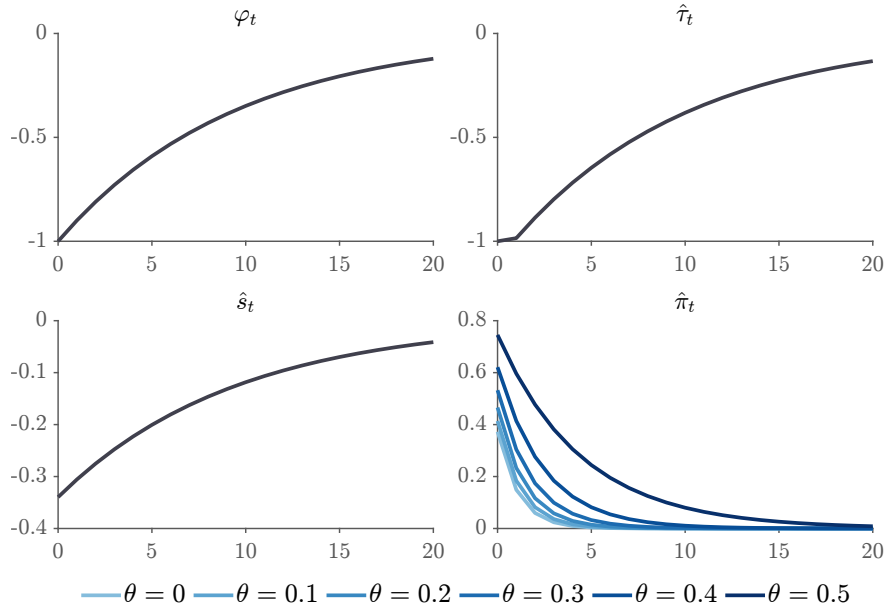


Figure 2: Dynamics of fiscally-led equilibrium in Ricardian model.

Figure 2 plots the equilibrium paths of the tax rate, the debt-to-GDP ratio, and the rate of inflation

in response to a deficit shock φ_t for different levels of inflation-indexed debt. We set $\phi = 0.4$ so that all values of θ shown remain within the fiscally-led region. As expected, the reaction of the tax rate and the debt-to-GDP ratio are independent of the share of inflation-indexed debt. The key difference lies in the observed rate of inflation: inflation-indexed debt causes additional loading on inflation under a fiscally-led policy mix. Higher equilibrium inflation is needed to ensure sufficient devaluation of the government debt stock, since only the *non-indexed* share of the debt stock can be devalued. The inflation that follows a deficit shock is increasing in the share of inflation-indexed debt when the fiscal authority does *not* promise a complete repayment of borrowing.

In sum, inflation-indexed debt in the simple model makes the policy space qualifying as a ‘fiscally-led policy mix’ smaller, but the consequences of being within that space become more detrimental: the inflationary pressure arising from an expansionary deficit shock is increasing in the share of outstanding inflation-indexed debt.

3 A non-Ricardian General Equilibrium Model with Indexed Debt

We proceed with a richer, albeit still somewhat tractable, general equilibrium framework based on [Angeletos et al. \(2024\)](#), who develop an OLG-NK model in the spirit of [Blanchard \(1985\)](#).⁵ The headline mechanism is a mortality friction that breaks Ricardian equivalence and can be considered as a general proxy for the liquidity risk in canonical heterogeneous-agent models. It allows for the analysis of purely distributive *transfer* shocks with relevant real and nominal effects, facilitating a clean analysis of the effects of inflation-indexed debt.

3.1 Model framework

As in the previous section, government bonds D_t are partially inflation-indexed, and can heuristically be decomposed into constant proportions of non-indexed and indexed debt:

$$D_t \equiv (1 - \theta)B_t + \theta\tilde{B}_t, \quad (8)$$

where θ is the time-invariant share of inflation-indexed debt. Inflation is zero in steady-state, with debt d_t , taxes t_t and household assets a_t measured in absolute deviations to allow for the possibility of zero-debt steady-states.

Households

The probability of the household surviving (i.e., the inverse of the mortality risk) from one period to another is $\omega \in (0, 1]$. Households are replaced whenever they die, keeping overall population levels constant. Conditional on being alive at date t , household i maximizes the present discounted value of its utility:

⁵[Rachel and Ravn \(2025\)](#) work in a similar framework.

$$\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k \left(\frac{C_{i,t+k}^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}} - \nu \frac{L_{i,t+k}^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}} \right), \quad (9)$$

where β is a common discount factor, ν scales the disutility of labour supply L_{it} , σ is the elasticity of intertemporal substitution, $1/\sigma$ is the coefficient of relative risk aversion, and φ is the Frisch elasticity of labour supply.

Households trade the partially-indexed government bond, D_t , earning a nominal rate of return R_t^D/ω if the household survives. In effect, there is a risk-free return of R_t^n/ω on the fraction $1 - \theta$ of debt that is not indexed and a return of $\frac{R_t^n}{\omega} \left(\frac{P_{t+1}}{P_t} \right)$ on the fraction θ that is indexed:

$$R_t^D = (1 - \theta)R_t^n + \theta R_t^n \left(\frac{P_{t+1}}{P_t} \right) = R_t^n \left(1 + \theta \left(\frac{P_{t+1}}{P_t} - 1 \right) \right). \quad (10)$$

The budget constraint of the household follows [Angeletos et al. \(2024\)](#), in that all households receive labour income $W_t L_{it}$ and dividends X_{it} , and face the same tax schedule T_{it} . Existing households make a contribution S_{it} to a social fund whose proceeds are distributed to newborn households.⁶ The social fund equalizes wealth across cohorts in steady state and thereby ensures that the steady-state real interest rate is $1/\beta$, but it does not eliminate the post-shock wealth effects generated by mortality risk. Letting $A_{i,t}^{nom}$ denote nominal financial wealth at the beginning of date t , the household-specific budget constraint is:

$$A_{i,t+1}^{nom} = \frac{R_t^D}{\omega} \left(A_{i,t}^{nom} + P_t \left(\underbrace{Y_{i,t}}_{\equiv W_t L_{i,t} + X_{i,t}} - C_{i,t} - T_{i,t} + S_{i,t} \right) \right). \quad (11)$$

We retain all other household-side assumptions from [Angeletos et al. \(2024\)](#). Dividends are identical across households i , labour supply is intermediated by unions to obtain $L_{it} = L_t$, and income and taxes faced by households are equalized.

The nominal portfolio return R_t^D is stochastic because it depends on realized inflation. The household's real return conditional on survival is given by

$$\frac{R_t^D}{\omega \Pi_{t+1}} = \frac{R_t^n}{\omega} \left(\frac{1 - \theta}{\Pi_{t+1}} + \theta \right). \quad (12)$$

Therefore, the exact consumption-saving first-order condition reads as follows:

$$C_{i,t}^{-\frac{1}{\sigma}} = \beta \mathbb{E}_t \left[C_{i,t+1}^{-\frac{1}{\sigma}} R_t^n \left(\frac{1 - \theta}{\Pi_{t+1}} + \theta \right) \right]. \quad (13)$$

The survival probability in preferences cancels the actuarially fair annuity premium $1/\omega$. Defining

⁶The transfers are $S_{it} = S^{new} = D^{SS} \geq 0$ and $S_{it} = S^{old} = -\frac{1-\omega}{\omega} D^{SS} \leq 0$, such that $(1 - \omega)S^{new} + \omega S^{old} = 0$.

the *ex ante* real interest rate through the Fisher equation as $r_t = r_t^n - E_t \pi_{t+1}$, a first-order approximation of equation (13) around the deterministic zero-inflation steady state gives the household-specific Euler equation

$$c_{i,t} = -\sigma(r_t + \theta E_t \pi_{t+1}) + E_t c_{i,t+1}. \quad (14)$$

The covariance between future marginal utility and the realized indexed payoff is second order around the deterministic steady state. This explains why expected inflation, rather than realized inflation, appears in the first-order Euler equation even though the exact return is stochastic.

The present-value budget constraint is obtained household by household from equation (13), the per-period budget constraint, and the household transversality condition. Let $\tilde{a}_{i,t}$ denote household i 's beginning-of-period real financial wealth inclusive of the social-fund payment. Linearizing and combining the household-specific present-value budget with equation (14) gives

$$\begin{aligned} c_{i,t} = & (1 - \beta\omega) \left(\tilde{a}_{i,t} + E_t \sum_{s=0}^{\infty} (\beta\omega)^s (y_{t+s} - t_{t+s}) \right) \\ & - \beta \left(\sigma\omega - (1 - \beta\omega) \frac{A^{SS}}{Y^{SS}} \right) E_t \left[\sum_{s=0}^{\infty} (\beta\omega)^s (r_{t+s} + \theta\pi_{t+1+s}) \right]. \end{aligned} \quad (15)$$

Income, taxes, and returns are common across households. Moreover, the social fund has a balanced budget and asset-market aggregation gives $\int_0^1 \tilde{a}_{i,t} di = a_t$. Integrating equation (15) over all cohorts therefore yields the aggregate generalization of the Permanent Income Hypothesis:

$$\begin{aligned} c_t = & (1 - \beta\omega) \left(a_t + E_t \sum_{s=0}^{\infty} (\beta\omega)^s (y_{t+s} - t_{t+s}) \right) \\ & - \beta \left(\sigma\omega - (1 - \beta\omega) \frac{A^{SS}}{Y^{SS}} \right) E_t \left[\sum_{s=0}^{\infty} (\beta\omega)^s (r_{t+s} + \theta\pi_{t+1+s}) \right]. \end{aligned} \quad (16)$$

a_t are net asset holdings of households and there is an additional inflation adjustment in the last term. This is the novelty relative to Angeletos et al. (2024), capturing income from inflation for holders of inflation-indexed debt.

Aggregate supply

The supply side of the model is standard and follows the canonical New Keynesian structure. The New Keynesian Phillips Curve is due to pricing frictions in the firm's problem:

$$\pi_t = \kappa y_t + \beta E_t \pi_{t+1}, \quad (17)$$

which, iterated forward, gives inflation as a function of current and future output gaps, that is, $\pi_t = \kappa \sum_{k=0}^{\infty} \beta^k E_t y_{t+k}$.

Government

The government budget constraint is standard, equating income to net expenditures:

$$D_{t+1} = R_t^D (D_t - P_t T_t),$$

where D_t is the stock of government debt and $R_t^D = R_t^n (1 + \theta(\Pi_{t+1} - 1))$ is the *ex post* return the government has to pay. The return is defined in terms of the *realised* rate of inflation, from t to $t + 1$.

Dividing both sides by P_t and denoting the real value of government debt by $D_t^R = D_t/P_t$,⁷ a linear approximation around a zero-inflation steady-state yields:

$$(D_{t+1}^R - D^{R,SS}) + D^{R,SS}(\Pi_{t+1} - 1) = R^{D,SS}[(D_t^R - D^{R,SS}) - (T_t - T^{SS})] + (D^{R,SS} - T^{SS})(R_t^D - R^{D,SS}).$$

Dividing both sides by steady-state output Y^{SS} and defining $d_t = (D_t^R - D^{R,SS})/Y^{SS}$ for debt and $t_t = (T_t - T^{SS})/Y^{SS}$ for taxes, the linearized budget constraint becomes:

$$d_{t+1} + \frac{D^{SS}}{Y^{SS}}\pi_{t+1} = \frac{1}{\beta}(d_t - t_t) + \frac{D^{SS}}{Y^{SS}}r_t^D,$$

where we simplify using the steady-state nominal return $R^{D,SS} = 1/\beta$ and the steady-state government budget constraint $D^{SS} = (D^{SS} - T^{SS})/\beta$.

The *ex post* return r_t^D is related to the risk-free nominal return r_t^n and *realised* inflation π_{t+1} through the inflation-indexation of the government bond, with the risk-free nominal return r_t^n itself related to the *ex ante* real return r_t and *expected* inflation $E_t\pi_{t+1}$ through the Fisher equation. As approximate log-deviations from steady state, the returns satisfy:

$$\begin{aligned} r_t^D &= r_t^n + \theta\pi_{t+1}, \\ r_t &= r_t^n - E_t\pi_{t+1}, \\ r_t^D &= r_t + E_t\pi_{t+1} + \theta\pi_{t+1}. \end{aligned}$$

The final expression for the government budget constraint is thus given by:

$$d_{t+1} = \frac{1}{\beta}(d_t - t_t) + \frac{D^{SS}}{Y^{SS}}r_t - \frac{D^{SS}}{Y^{SS}}((1 - \theta)\pi_{t+1} - E_t\pi_{t+1}). \quad (18)$$

The novelty here is the multiplication of *realised* future inflation by $(1 - \theta)$, which is the share of non-indexed debt. Intuitively, inflation-indexed debt cannot be devalued through surprise inflation. The ability of governments to devalue their debt in real terms is therefore constrained by indexed debt, with implications for the evolution of the debt stock in the presence of inflation.⁸

⁷The resulting government budget constraint is $D_{t+1}^R \Pi_{t+1} = R_t^D (D_t^R - T_t)$.

⁸We furthermore impose the no-Ponzi condition of Angeletos et al. (2024), i.e., $E_t \lim_{T \rightarrow \infty} \beta^T d_{t+T} = 0$, which arises from

The steady-state of the model requires $x_{-1} = 0 \forall x \in \{d, t, r, y, \pi\}$. Starting from the steady state, equation (18) defines the initial change in the debt stock as a function of surprise inflation:

$$d_0 = -\frac{D^{SS}}{Y^{SS}}(1 - \theta)\pi_0,$$

and the presence of inflation-indexed debt generally reduces the initial reaction of the value of the debt stock to inflationary pressure.

For now, we close the model with the monetary and fiscal and policy rules in Angeletos et al. (2024), although we already want to highlight that, later on, these rules will be relaxed substantially:

$$r_t = \phi y_t, \quad (19)$$

$$t_t = -\varepsilon_t + \tau_d(d_t + \varepsilon_t) + \tau_y y_t. \quad (20)$$

3.2 Aggregate demand

Aggregate demand is derived by imposing market clearing conditions $a_t = d_t$ and $y_t = c_t$ and substituting the monetary policy rule (19) into the intertemporal budget constraint (16):

$$y_t = (1 - \beta\omega) \left(d_t + \mathbb{E}_t \sum_{s=0}^{\infty} (\beta\omega)^s \left[\left(1 - \frac{\Gamma\phi}{1 - \beta\omega} \right) y_{t+s} - t_{t+s} - \frac{\Gamma\theta}{1 - \beta\omega} \pi_{t+1+s} \right] \right), \quad (21)$$

where $\Gamma \equiv \beta \left(\sigma\omega - (1 - \beta\omega) \frac{D^{SS}}{Y^{SS}} \right)$.

In recursive form, the aggregate demand relationship simplifies to:

$$\mathbb{E}_t y_{t+1} - \frac{\Gamma\theta}{\beta\omega} \mathbb{E}_t \pi_{t+1} - (1 - \beta\omega) \mathbb{E}_t d_{t+1} = \frac{\beta\omega(1 - \tau_y) + \Gamma\phi + \tau_y}{\beta\omega} y_t - \frac{(1 - \beta\omega)(1 - \tau_d)}{\beta\omega} (d_t + \varepsilon_t), \quad (22)$$

where taxes have been substituted out using fiscal policy rule (20).

Equation (22) generalizes the aggregate demand equation to account for inflation-indexed debt. Through θ , inflation indexation weakens the relationship between current output and future inflation. Together with the New Keynesian Phillips Curve (17) and the government debt valuation equation (18), it gives a full characterization of the model's dynamics.

3.3 Equilibrium

The definition of the *competitive equilibrium* is standard.

Definition 1 A competitive equilibrium is a series $\{c_t, y_t, \pi_t, a_t, d_t, t_t, r_t\}_{t=0}^{\infty}$ that satisfies the aggregate demand function, the New Keynesian Phillips Curve, market clearing, the government's flow budget constraint, the no-Ponzi condition, and the monetary and fiscal policy rules.

rational optimizing behaviour on behalf of the households.

Without mortality risk

We begin by analysing a simplified version of the model, abstaining from mortality risk by setting $\omega = 1$. This environment, which is similar to a representative agent New Keynesian (RANK) model, retains the fundamental block recursivity of the system, whereby the dynamics of output and inflation are independent of the dynamics of debt. To see this, substitute the government budget constraint (18) with the policy rules (19) and (20) into the aggregate demand equation (22) under $\omega = 1$:

$$\mathbb{E}_t y_{t+1} - \theta \sigma \mathbb{E}_t \pi_{t+1} = (1 + \sigma \phi) y_t, \quad (23)$$

The output-inflation block in the absence of mortality risk is therefore independent of the evolution of government debt, even with indexed debt (Kaplan, 2025; Rachel and Ravn, 2025). This mirrors the canonical formulations of RANK models, except for the slight indexed-debt driven adjustment. The above equation combines with the New Keynesian Phillips Curve (17) and the government budget constraint (18) to define the first-order system:

$$\mathbb{E}_t \begin{bmatrix} y_{t+1} \\ \pi_{t+1} \\ d_{t+1} \end{bmatrix} = \begin{bmatrix} 1 + \phi \sigma - \frac{\kappa \sigma \theta}{\beta} & \frac{\sigma \theta}{\beta} & 0 \\ -\frac{\kappa}{\beta} & \frac{1}{\beta} & 0 \\ \frac{D^{SS}}{Y^{SS}} \phi - \frac{\tau_y}{\beta} - \frac{D^{SS}}{Y^{SS}} \frac{\kappa \theta}{\beta} & \frac{D^{SS}}{Y^{SS}} \frac{\theta}{\beta} & \frac{1}{\beta} (1 - \tau_d) \end{bmatrix} \begin{bmatrix} y_t \\ \pi_t \\ d_t + \varepsilon_t \end{bmatrix}. \quad (24)$$

The system is block recursive for all $\theta \geq 0$, so inflation indexation of debt in itself does not overcome the result that government debt is block-exogenous with respect to output and inflation in RANK models (Leeper, 1991; Angeletos et al., 2024). The policy space that supports a unique saddle-path stable equilibrium will therefore still split into areas conventionally labelled 'active' and 'passive' with respect to the actions of each policymaker.

The system has one predetermined variable d_t and two forward-looking variables (y_t, π_t) , so determinacy requires one eigenvalue inside and two eigenvalues outside the unit circle. In the fiscally-led equilibrium, the eigenvalue $\beta^{-1}(1 - \tau_d)$ associated with the government budget constraint must be outside the unit circle, which requires $\tau_d < 1 - \beta$. The conditions under which the other two eigenvalues straddle the unit circles in the fiscally-led equilibrium are as follows:

Proposition 1 *To ensure a unique saddlepath equilibrium when $\omega = 1$ under a fiscally-led policy mix $\tau_d < 1 - \beta$, the restriction on ϕ , the monetary policy parameter, tightens with the degree of inflation-indexed debt. The formal restriction for a fiscally-led equilibrium in the absence of mortality risk is:*

$$\max \left\{ -\frac{1}{\sigma}, -\frac{1}{\sigma} \left[2 - \frac{\kappa \sigma \theta}{1 + \beta} \right] \right\} < \phi < -\frac{\kappa \theta}{1 - \beta}. \quad (25)$$

Proof. See appendix B.1. ■

The conditions on the monetary policy parameter ϕ are tighter for determinacy in the fiscally-led regime with a positive share of inflation-indexed debt, reminiscent of our result in section 2. A

larger share θ of indexed debt, or a steeper Phillips curve (higher κ) all push the right-hand bound in (25) leftward, so increased monetary passivity is needed to sustain fiscal dominance. Otherwise, there are multiple equilibria with self-fulfilling fluctuations. The bounds on ϕ for monetary dominance can also tighten when the share of indexed debt is sufficiently high.⁹ Figure 3 depicts Proposition 1, without indexed debt in the left panel and with indexed debt ($\theta = 0.1$) in the right panel. The affected combinations for a unique monetary-led equilibrium are not restricted by the presence of inflation-indexed debt (the top-right green area stays the same in the left and right panels),¹⁰ but any fiscally-led equilibrium can only be sustained by increased degrees of monetary passivity (the bottom-left green area gets smaller).

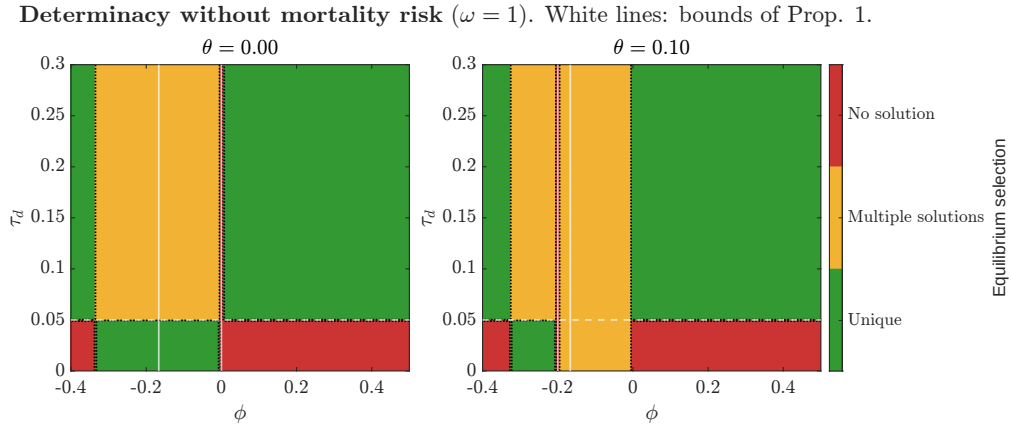


Figure 3: The determinacy properties of the RANK model.

3.4 With mortality risk

We now explore the more general model with mortality risk. As we will see, debt is not block-exogenous with respect to output and inflation here, inducing real effects of fiscal policy except in some edge cases, making fiscal policy therefore generally non-Ricardian. The first-order system for the joint dynamics of $\{y_t, \pi_t, d_t\}$ with $\omega \neq 1$ is:

$$\begin{pmatrix} 1 & -\frac{\Gamma\theta}{\beta\omega} & -(1-\beta\omega) \\ 0 & 1 & 0 \\ 0 & -\theta\frac{D^{SS}}{Y^{SS}} & 1 \end{pmatrix} E_t \begin{pmatrix} y_{t+1} \\ \pi_{t+1} \\ d_{t+1} \end{pmatrix} = \begin{pmatrix} \frac{\beta\omega(1-\tau_y)+\Gamma\phi+\tau_y}{\beta\omega} & 0 & -\frac{(1-\beta\omega)(1-\tau_d)}{\beta\omega} \\ -\frac{\kappa}{\beta} & \frac{1}{\beta} & 0 \\ \frac{D^{SS}}{Y^{SS}}\phi - \frac{\tau_y}{\beta} & 0 & \frac{1}{\beta}(1-\tau_d) \end{pmatrix} \begin{pmatrix} y_t \\ \pi_t \\ d_t + \varepsilon_t \end{pmatrix}. \quad (26)$$

Multiplying by the inverse of the left-hand matrix defines a system in the usual form:

$$E_t \begin{pmatrix} y_{t+1} \\ \pi_{t+1} \\ d_{t+1} \end{pmatrix} = A \begin{pmatrix} y_t \\ \pi_t \\ d_t + \varepsilon_t \end{pmatrix}. \quad (27)$$

⁹See Appendix B.2 for technical details.

¹⁰Clearly, this can also be tightened for extremely high levels of the EIS θ , the slope of the NKPC κ , and high shares of indexed debt θ .

The characteristic polynomial of A is relatively complex. It is nonetheless possible to provide some characterization of the factors that contribute to saddle-path stability, as well as to translate this characterization into parametric bands in the (ϕ, τ_d) -space. The details are in Appendix B.3.

Proposition 2 *Suppose that $\omega \in (0, 1)$, $\tau_y > 0$, $\tau_d \in [0, 1]$, $\phi > -1/\sigma$, and*

$$\mathcal{J} \equiv (1 - \beta\omega)(1 - \frac{1}{\omega}) < 0, \quad \delta \equiv \sigma + \frac{D^{SS}}{Y^{SS}}\mathcal{J} > 0, \quad \theta < \frac{1 + \beta}{\kappa\sigma}.$$

Let $p(\lambda) = \det(\lambda I - A)$, define

$$C(\tau_d) \equiv \sigma(\tau_d - (1 - \beta)) + \beta\mathcal{J} \frac{D^{SS}}{Y^{SS}},$$

and let

$$\Delta(\phi, \tau_d) \equiv 1 - S_2(A) + \text{tr}(A) \det(A) - \det(A)^2,$$

where $S_2(A)$ is the sum of the principal 2×2 minors of A . Then $p(-1) < 0$, and the model has a unique saddle path-stable equilibrium if and only if

$$C(\tau_d) [\theta\kappa + (1 - \beta)\phi] > \tau_y \mathcal{J} (1 - \beta), \quad (28)$$

together with either

$$\Delta(\phi, \tau_d) < 0, \quad (29)$$

or

$$\Delta(\phi, \tau_d) > 0 \quad \text{and} \quad |\text{tr}(A)| > 3. \quad (30)$$

The first condition can be written as a monetary-policy boundary. Let

$$\tau_d^0 \equiv 1 - \beta - \frac{\beta\mathcal{J}}{\sigma} \frac{D^{SS}}{Y^{SS}}, \quad \phi^*(\tau_d) \equiv \frac{\tau_y \mathcal{J}}{C(\tau_d)} - \frac{\theta\kappa}{1 - \beta}.$$

Then equation (28) requires $\phi < \phi^*(\tau_d)$ when $\tau_d < \tau_d^0$ and $\phi > \phi^*(\tau_d)$ when $\tau_d > \tau_d^0$. Moreover, for

$$\tau_d^\# \equiv 1 - \left[\beta^2 + \frac{\beta^2}{\sigma} \frac{D^{SS}}{Y^{SS}} \mathcal{J} \right],$$

$\Delta(\phi, \tau_d)$ is strictly convex in ϕ when $\tau_d^\# < \tau_d < 1$. Whenever the equation $\Delta(\phi, \tau_d) = 0$ has two distinct real solutions in ϕ , equation (29) is equivalent to a band $\phi_-(\tau_d; \theta) < \phi < \phi_+(\tau_d; \theta)$, subject also to equation (28). The remaining determinate parameterizations, when present, are characterized by the upper-wedge condition (30).

Proof. See appendix B.3. ■

The assumption $\phi > -1/\sigma$ is a standard restriction on the monetary policy parameter that allows us to look only at what is conventionally deemed the empirically relevant part of the space of possible

monetary policy rules. $\mathcal{T} = (1 - \beta\omega)(1 - \frac{1}{\omega}) < 0$ is always true, while $\mathcal{S} = \sigma + \frac{D^{SS}}{\gamma^{SS}}\mathcal{T} > 0$ is also usually not restrictive. The Proposition describes, in effect, how the bands within which saddle path-stability is ensured are moving under the presence of indexed debt. In particular, *under the assumed functional forms for monetary and fiscal policy rules*, indexed debt usually limits the degree to which non-stabilizing fiscal rules can induce saddle path-stability, taking the monetary policy rule parametrization as given.

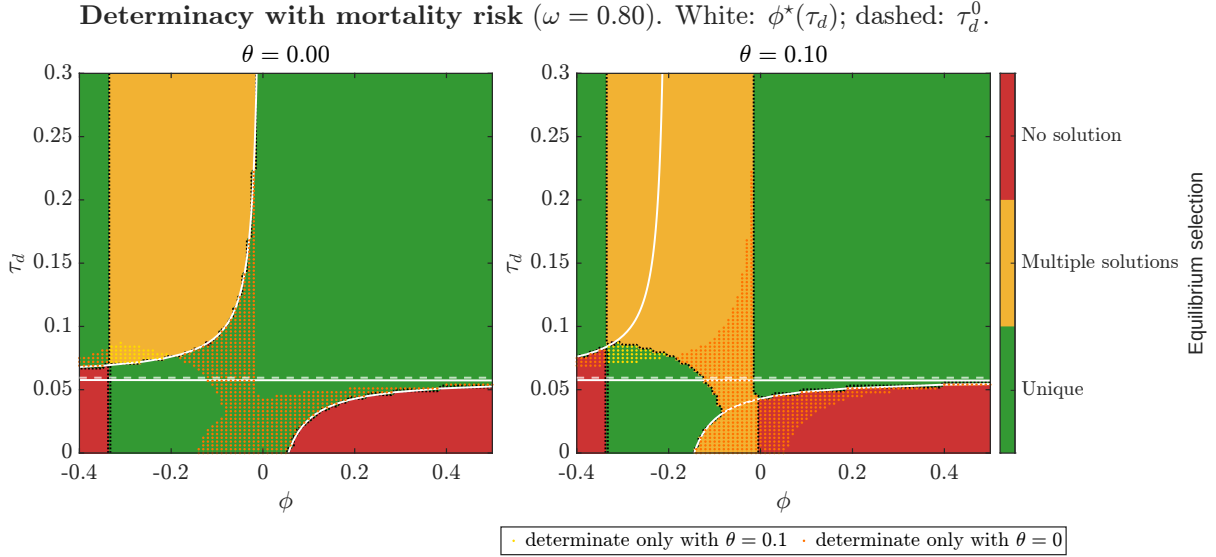


Figure 4: Visualization of Proposition 2.

Figure 4 visualizes Proposition 2, showing parameterizations of fiscal policy τ_d and monetary policy ϕ for which the first-order system with mortality risk has a unique saddlepath solution. The left panel depicts the policy space without inflation-indexed debt, replicating the result in Angeletos et al. (2024) and Rachel and Ravn (2025) that the presence of mortality risk creates a continuous policy space of saddle path-stable equilibria with no clear distinction between so-called fiscally and monetary-led equilibria. The right panel of Figure 4 depicts the same system when inflation-indexed debt is 10% of the overall government debt stock.¹¹ In line with Proposition 2, the green area shrinks at low values of τ_d and a lower monetary policy parameter ϕ is needed to support a unique saddlepath equilibrium.

What is the intuition behind this result? Inflation-indexed debt can be thought of as a type of creditor-side inflation hedge. For a given expansionary, deficit-financed fiscal measure, for instance, the presence of inflation-indexed debt should stabilize the value of that part of the government debt stock, as such debt retains its real value from the perspective of the household. However, when a fiscal policy authority commits to being *very* expansionary ($\tau_d \searrow 0$), we run into a scenario where self-fulfilling fluctuations are possible if the *monetary* authority in turn engages in interest rate policies by which real interest rates remain approximately stable.

¹¹This is above the share of TIPS in the stock of US Federal Debt, which is about 7.5%, but below the share of inflation-linked Gilts in the stock of UK Government debt, which is around 25%.

To see this, start from considering the fiscal policy authority. When it reacts insufficiently by appropriate taxation to changes of the real value of debt from equilibrium, it is possible that the raised additional taxes are *insufficient* to serve the additional face value of maturing inflation-indexed debt in response to an expansionary fiscal shock. When the monetary authority intends to keep real rates at least approximately stable, this problem compounds for the fiscal authority as the overall debt service cost also does not decrease. Through the combination of the presence of indexed debt, very expansionary fiscal policy, and a monetary authority that intends to ‘fight back’ against inflationary pressure, it is then possible for self-fulfilling fluctuations to materialize.

Under such a rich system, characterizing inflation analytically becomes difficult. We are nonetheless interested in the effect of inflation-indexed debt, and therefore perform the following experiment: for the entirety of the (relevant) policy space, we calculate the one-year change in the price level in response to a 1% deficit-to-GDP shock under two scenarios: once setting the share of inflation-indexed debt to 10% ($\theta = 0.1$), and once without any inflation-indexed debt ($\theta = 0$). Then, we take the difference of the two observed inflation rates, allowing us to characterize the impact of inflation-indexed debt on price level dynamics in the model. Figure 5 shows the results of this exercise, varying the core fiscal reaction parameter τ_d on the y-axis and the monetary policy parameter ϕ on the x-axis.

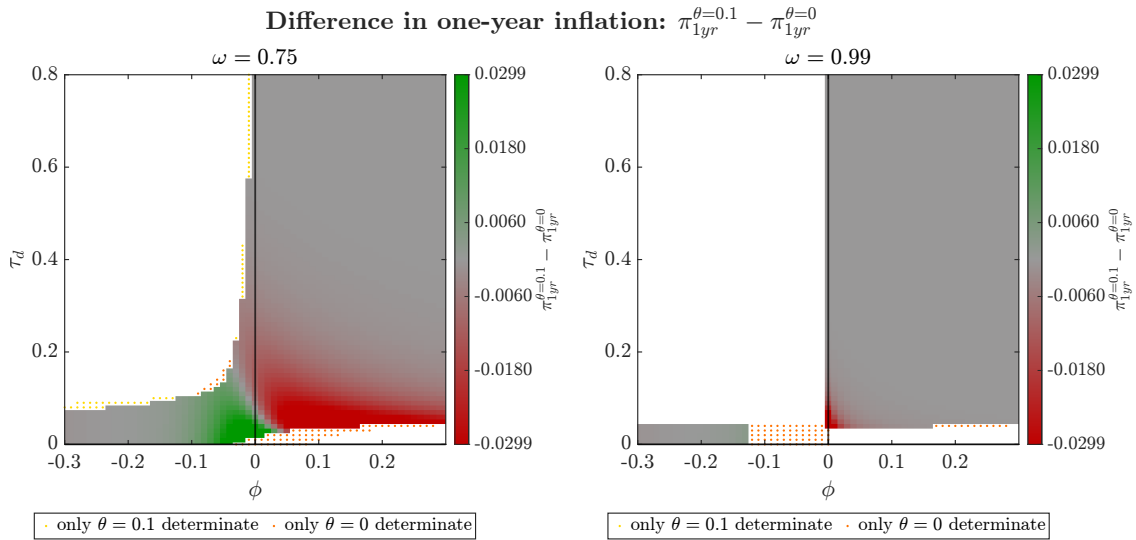


Figure 5: Difference in one-year inflation rates due to indexed debt in response to a 1% deficit-to-GDP shock in dependence on the policy parameters.

While the clear-cut distinction between fiscally-led and monetary-led policy mixes has become seemingly difficult under the presence of mortality risk, as clear from figure 4 and the interpretation in [Rachel and Ravn \(2025\)](#), figure 5 shows a very interesting novel result that arises due to the presence of inflation-indexed debt: we have one clear area in which inflation-indexed debt has a *net positive* impact on inflation, and one clear area in which it has a *net negative* impact on inflation,

i.e., an area where inflation-indexed debt can be *disinflationary*.¹² Low values of τ_d and ϕ are increasing the propensity of indexed debt to be inflationary, which can therefore be seen as a tool in helping measure *approximate fiscal dominance*, even when no clearly distinct policy areas persist.

How is it possible for inflation-indexed debt to induce slight *disinflationary* pressure in response to a deficit shock? This holds particularly true for high values of ϕ , i.e., for monetary reaction parameters that induce a real rate increase in equilibrium. In such circumstances, the real value of inflation-indexed debt is *increasing* once a deficit shock materializes through the promise of the central bank to crush additional inflationary pressure arising from the deficit shock, yielding a surprise *positive* revaluation of the outstanding indexed debt stock. Then, the overall value of the stock of government debt is increasing through the presence of inflation-indexed debt, yielding some disinflationary pressure in equilibrium. The presented response of equilibrium inflation in dependence on the share of inflation-indexed debt also rationalizes the result in figure 4 to some extent, as the overall volatility of the price level is clearly most pronounced through the presence of indexed debt for low values of τ_d , irrespective of the monetary reaction parameter, which mostly influences the *direction* of the observed price level volatility.

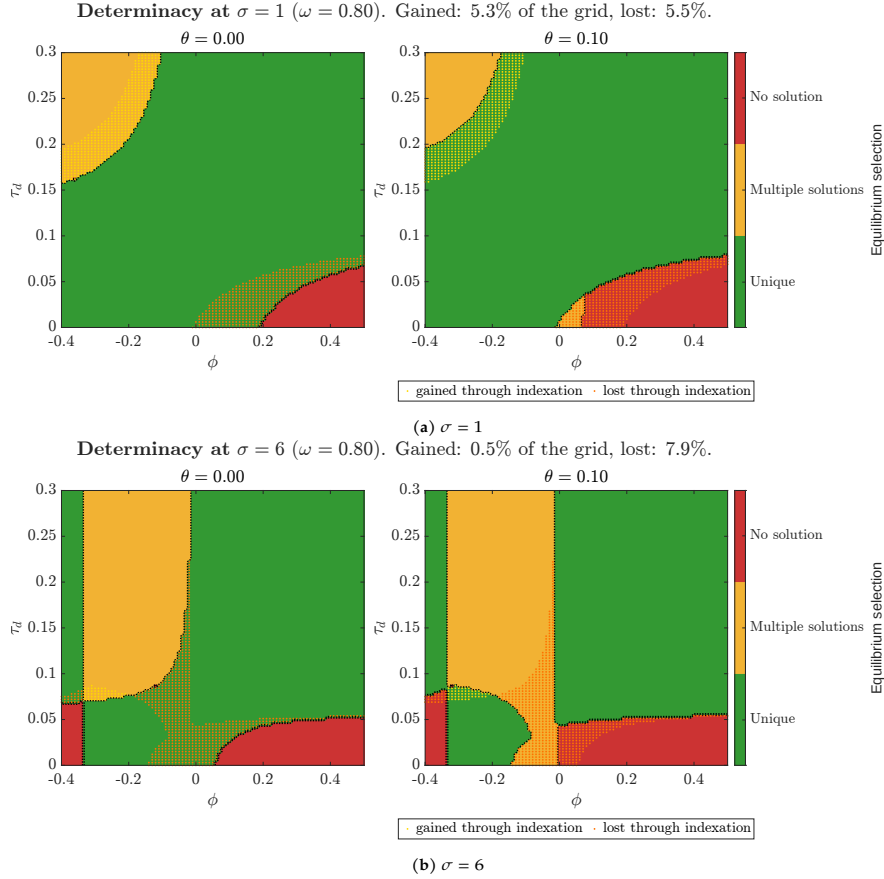


Figure 6: The effect of the elasticity of intertemporal substitution on the model, contrasting the cases of $\sigma = 1$ and $\sigma = 6$.

¹²This was not the case in the RANK model, where inflation-indexed debt was either inflationary (under a fiscally-led policy mix), or had no impact on the unique solution $\pi_t = 0$ (in the monetary-led policy mix).

Relative to proposition 2, figure 6 sheds additional light on the effect of intertemporal substitution. Under the preference specification used here, a higher value of σ raises the elasticity of intertemporal substitution and lowers relative risk aversion, which is given by $1/\sigma$. In particular, the relatively even-sided effect on determinacy properties caused by the presence of indexed debt becomes increasingly *one-sided* as the elasticity of intertemporal substitution goes up: for higher levels of σ , inflation-indexed debt still reduces the feasible policy space for activist fiscal policy. However, it does *not* increase the policy space under relatively passive fiscal policy in the parameterization shown. Therefore, for high levels of the elasticity of intertemporal substitution, inflation-indexed debt generally only reduces the feasible policy space in this numerical experiment.

4 Inflation-indexed debt as a joint fiscal-monetary instrument

In the previous section, we restricted ourselves to relatively simple policy rules that were broadly in line with Angeletos et al. (2024) and Rachel and Ravn (2025). However, it is logical for the presence of inflation-indexed debt to alter the conduct of policymakers: both monetary and fiscal policy might anticipate some of the effects of indexed debt that we just explored, internalizing them in their own policymaking. We here explore such alternative policy rules, showing how the shift in the determinate space can be unwound, or even inverted, with appropriate policy measures.

4.1 Fiscal and monetary rules accounting for the presence of indexed debt

In section 2, we specified the monetary rule as equation (19) and the fiscal rule as equation (20). However, the presence of inflation-indexed debt can plausibly be accounted for by either policy authority, especially in the light of the possibly detrimental consequences in terms of equilibrium (in)existence or (dis)inflationary pressure.

It is possible to create a myriad of such policy specifications. Here, we focus on one particular instance, comprised of a monetary authority that aims to mitigate the impact of inflation-indexed debt on the aggregate demand equation, paired with a fiscal authority that is aware of the additional cost of serving such indexed debt upon the realization of a positive inflationary shock.

The alternative monetary rule: following equation (16), inflation-indexed debt manifests in the household policy function by inducing a direct wedge on equilibrium real interest rates. A monetary authority tasked with providing broad economic stabilization might then prefer to nullify this effect to the degree possible within its mandate. In this case, the central bank can simply follow a real rate rule that internalizes the additional pressure on the portfolio-adjusted real rate earned by the household. Formally:

$$r_t = \phi y_t - \theta \mathbb{E}_t \pi_{t+1} \quad (31)$$

after log-linearization. The novel term is given by $-\theta \mathbb{E}_t \pi_{t+1}$. Under such a monetary rule, the pol-

icy rate set is effectively a *portfolio-adjusted real rate*, which acts as a stabilizing tool on the aggregate demand side in the presence of unexpected inflationary shocks.

Fiscal policy, in turn, can try to internalize anticipated expenditures arising from the presence of such debt. In that case, fiscal policy can simply be specified to follow:

$$t_t = -\varepsilon_t + \tau_d(d_t + \varepsilon_t) + \tau_y y_t + \frac{\beta\omega}{1-\beta\omega} \frac{D^{SS}}{Y^{SS}} \theta \tau_\pi \mathbb{E}_t \pi_{t+1} \quad (32)$$

after log-linearization. Here, the novel term in the rule is $\frac{\beta\omega}{1-\beta\omega} \frac{D^{SS}}{Y^{SS}} \theta \tau_\pi \mathbb{E}_t \pi_{t+1}$, with the novel policy parameter τ_π . The terms $\frac{\beta\omega}{1-\beta\omega} \frac{D^{SS}}{Y^{SS}} \theta$ weight this novel reaction parameter in line with intertemporal discounting and the overall outstanding indexed debt stock. Factually, such a fiscal rule ensures that a fiscal policy authority expecting increased outlays for inflation-indexed debt attempts to raise additional funds through taxation to cover such expenses, which makes it an indexed-debt-contingent fiscal feedback rule that was chosen to internalize the model's non-Ricardian wedge. Conditional on the return rates for either assets being equivalent in real terms, *expected* inflationary pressure would then even yield a windfall gain on behalf of the fiscal authority.

With these two rules, the aggregate demand equation (22) changes to the following specification:

$$\mathbb{E}_t y_{t+1} - \tau_\pi \frac{D^{SS}}{Y^{SS}} \theta \mathbb{E}_t \pi_{t+1} - (1-\beta\omega) \mathbb{E}_t d_{t+1} = \frac{\beta\omega(1-\tau_y) + \Gamma\phi + \tau_y}{\beta\omega} y_t - \frac{(1-\beta\omega)(1-\tau_d)}{\beta\omega} (d_t + \varepsilon_t). \quad (33)$$

Future inflation continues to matter for household aggregate demand, but now only through the distortionary taxation induced by the fiscal authority in proportion to its anticipated additional cost of serving maturing indexed debt. The initial 'base' effect through a wealth channel induced by a mismatch between the real interest rate set by the central bank and the factual portfolio return has been nullified.

The third equilibrium condition, the intertemporal debt valuation equation (initially given by equation (18)), similarly changes after taking expectations and inserting our novel tax rule to

$$\mathbb{E}_t d_{t+1} + \frac{\omega}{1-\beta\omega} \frac{D^{SS}}{Y^{SS}} \theta \tau_\pi \mathbb{E}_t \pi_{t+1} = \left(\frac{D^{SS}}{Y^{SS}} \phi - \frac{\tau_y}{\beta} \right) y_t + \frac{1-\tau_d}{\beta} (d_t + \varepsilon_t). \quad (34)$$

The direct wedge arising from the indexed pecuniary return has vanished due to the adjustment of the monetary rule. Expected inflation, however, continues to affect the fiscal balance sheet through the taxation adjustment, now with the coefficient $\omega \frac{D^{SS}}{Y^{SS}} \theta \tau_\pi / 1 - \beta\omega$.

Summarizing equations (33) and (34), together with the NKPC (17), gives us the following 3×3 first-difference system:

$$\mathbb{E}_t \begin{bmatrix} y_{t+1} \\ \pi_{t+1} \\ d_{t+1} \end{bmatrix} = \begin{bmatrix} C_y^{alt} + (1 - \beta\omega)G_y^{alt} - \frac{\kappa \frac{D^{ss}}{Y^{ss}} \theta \tau_\pi (1 - \omega)}{\beta} & \frac{\frac{D^{ss}}{Y^{ss}} \theta \tau_\pi (1 - \omega)}{\beta} & \frac{1 - \beta\omega}{\beta} (1 - \tau_d) \left(1 - \frac{1}{\omega}\right) \\ -\frac{\kappa}{\beta} & \frac{1}{\beta} & 0 \\ G_y^{alt} + \frac{\omega \frac{D^{ss}}{Y^{ss}} \theta \tau_\pi \kappa}{1 - \beta\omega} \frac{1}{\beta} & -\frac{\omega \frac{D^{ss}}{Y^{ss}} \theta \tau_\pi}{(1 - \beta\omega)\beta} & \frac{1}{\beta} (1 - \tau_d) \end{bmatrix} \begin{bmatrix} y_t \\ \pi_t \\ d_t + \varepsilon_t \end{bmatrix}, \quad (35)$$

where $C_y^{alt} \equiv \frac{\beta\omega(1-\tau_y)+\Gamma\phi+\tau_y}{\beta\omega}$, $G_y^{alt} \equiv \frac{D^{ss}}{Y^{ss}}\phi - \frac{\tau_y}{\beta}$. The coefficient on expected inflation in the first row reflects the difference between the household demand-side loading $\frac{D^{ss}}{Y^{ss}}\theta\tau_\pi$ and the fiscal loading $(1 - \beta\omega)(\omega \frac{D^{ss}}{Y^{ss}}\theta\tau_\pi / (1 - \beta\omega)) = \omega \frac{D^{ss}}{Y^{ss}}\theta\tau_\pi$. In the representative-agent limit $\omega = 1$, these two effects cancel from the output-inflation block.

Determinacy under baseline and novel policy rules ($\omega = 0.80$, $\sigma = 1$, $\tau_\pi = 1.0$)

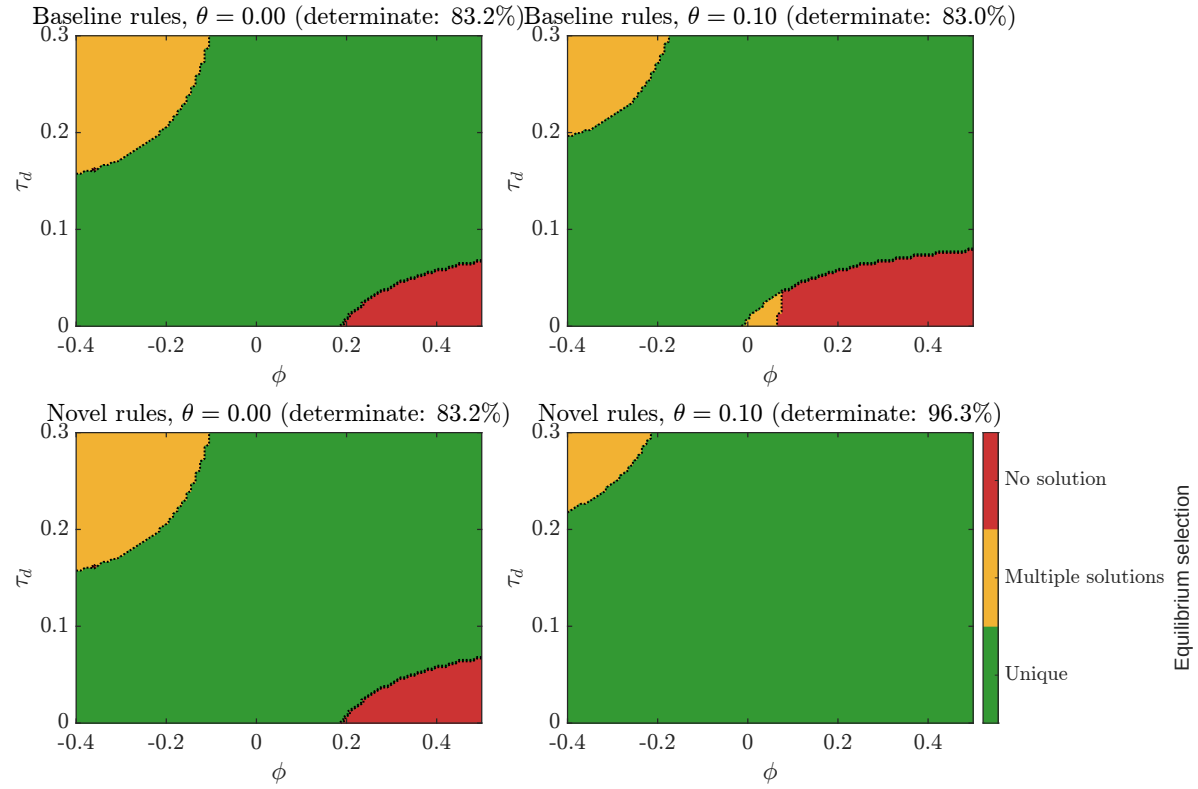


Figure 7: The determinacy properties of the model with mortality risk in dependence on the presence of indexed debt for the novel monetary and fiscal rules (31) and (32). The novel policy parameter τ_π has been set to 1.0.

Figure 7 depicts the impact of these novel rules on the determinacy properties of the system. In short, changing the fiscal and monetary rules to reflect a partial internalization of the anticipated effects of indexed debt *widens* the space under which saddle path-stable equilibria are possible unilaterally. In particular, even for relatively low values of τ_d , which indicate fiscal authorities

committing to very expansionary policy measures, determinate equilibria are possible even for central bank policies that are tightening the economy in the presence of high inflation rates. In this scenario, the presence of indexed debt allows a foray into the area conventionally labelled as “active/active”, which has thus far not been possible in other models, despite its possible real-world relevance in the light of the post-Covid inflationary episode.

Intuitively, the presence of the novel term in the fiscal rule facilitates more ‘active’ fiscal rules, since the presence of indexed debt does not weigh negatively on the fiscal budget constraint, since additional outlays for inflation-indexed debt are covered by the novel ‘inflation stabilizer’ in the fiscal rule (32).

To complement the analysis, we now also look at is the impact of the novel rules on inflationary pressure in the face of deficit shocks, shown by figures 8 and 9.

When the monetary rule absorbs the household-level windfall gain arising from their holdings of indexed debt in the presence of surprise inflation, the presence of indexed debt does *not* induce any additional inflationary pressure, and, by that same logic, the government’s intertemporal budget constraint does not deteriorate through the presence of additional indexed debt. If anything, it is possible for inflation-indexed debt to depress inflationary pressure in response to deficit shocks when the government receives an additional windfall gain through its taxation in the wake of expected inflation. Note that, of course, the inflationary pressure in response to the deficit shock remains positive and particularly elevated for low levels of τ_d , but the change in the fiscal and monetary rules simply allows for a reduction in the magnitude of that pressure when indexed debt is present. Simply put, the *additional* inflationary pressure coming through indexed debt can be nullified, but nothing more.¹³

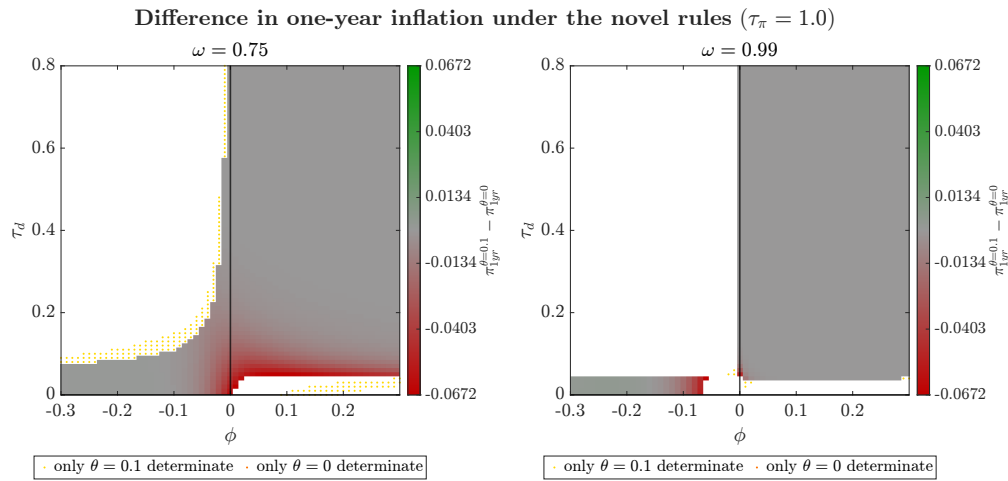


Figure 8: The difference in one-year inflation in the model with mortality risk between the cases $\theta = 0.1$ and $\theta = 0$ with fiscal rules (31) and (32). The novel policy parameter τ_π has been set to 1.0.

¹³Appendix B.4 zooms into the effects of the novel fiscal rule parameter τ_π , which captures the tax feedback to indexed debt payments and which was set to 1 so far. In particular, we establish numerically there that its effect on both the determinacy properties and the inflationary pressure arising from a fiscal impulse are fairly small. Thus, our results in this section are especially driven by the change to the monetary rule, and less so by changes to the fiscal rule.

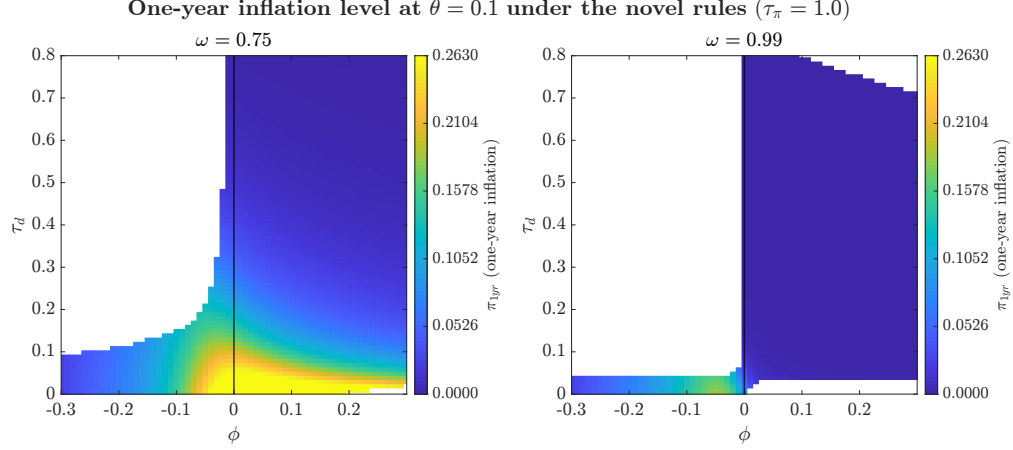


Figure 9: The level of one-year inflation rates with inflation-indexed debt ($\theta = 0.1$) under the policy rules (31) and (32). The novel policy parameter τ_π has been set to 1.0.

4.2 Adding simple CB balance sheet policy

The preceding exercises treat the share of inflation-indexed debt as an object chosen by the fiscal authority, with all such debt being held by the private sector. This is a useful benchmark, but it abstracts from a potentially important feature of actual policy: central banks can change the private exposure to indexed debt by purchasing indexed government bonds and issuing nominal reserves. Such a policy does not change the total amount of Treasury debt outstanding, however, it changes the composition of assets held by households. In its simplest form, it swaps an inflation-insured claim for a nominal claim. We now look at the effects of such policy, distinguishing the share of indexed debt in the total debt stock issued by the treasury from the respective shares held by the central bank and households in our model.

Let total Treasury debt still be denoted by D_t . The Treasury issues a fixed share $\theta^T \in [0, 1]$ of this debt in inflation-indexed form. The central bank holds a share $\theta_t^{cb} \in [0, \theta^T]$ of total Treasury debt in indexed bonds, financing these holdings one-for-one with nominal reserves. The household sector holds the residual indexed share θ_t^h . Therefore, $\theta^T = \theta_t^h + \theta_t^{cb}$.

Of course, we need to specify the central bank balance sheet and the way the split is determined in greater detail. For now, we assume that central bank liabilities solely consist of interest-bearing reserves, which are remunerated at the risk-free nominal rate R_t^n , i.e., at the same rate as non-indexed government bonds (Curdia and Woodford, 2011; Reis, 2017). Reserves and nominal bonds are thus perfect substitutes in pecuniary terms. The central bank generally operates with zero net worth and remits its net income to the Treasury each period. Its balance sheet is then given by:

$$\underbrace{\theta_t^{cb} D_t}_{\text{assets: indexed bonds}} = \underbrace{V_t}_{\text{liabilities: reserves}}, \quad (36)$$

where V_t denotes nominal reserves. Between t and $t+1$, the central bank earns the indexed cash return on its assets and pays R_t^n on its reserve liabilities, and the difference is remitted to the Treasury.

Furthermore, and reminiscent of the baseline model, households hold the composite portfolio that the consolidated government supplies, so there is no separate market in which they can re-trade its composition. Under these assumptions, the Treasury's liabilities are $(1 - \theta^T)D_t$ in nominal and $\theta^T D_t$ in indexed bonds, the central bank holds $\theta_t^{cb} D_t$ of the latter against reserves $V_t = \theta_t^{cb} D_t$, and household financial wealth is

$$P_t A_t = \underbrace{(1 - \theta^T)D_t}_{\text{nominal bonds}} + \underbrace{\theta_t^h D_t}_{\text{indexed bonds}} + \underbrace{V_t}_{\text{reserves}} = (1 - \theta^T)D_t + (\theta^T - \theta_t^{cb})D_t + \theta_t^{cb} D_t. \quad (37)$$

Hence, the market-clearing condition $a_t = d_t$ survives unchanged, and so do the steady state and the composite coefficient Γ , since $A^{SS} = D^{SS}$ is unaffected by the composition of the portfolio. Differently put, central-bank purchases of indexed bonds do not create net private wealth, they just replace privately held indexed bonds with nominal reserves. The entire stock of aggregate wealth in the system is thus unchanged, but households do not capture the full exposure to their associated benefits and risks.

For the following, it will be useful to define the central bank's deviation from its steady-state indexed-bond target, which will help us to characterize the local dynamics. Define therefore $\bar{\theta}^{cb} \in [0, \theta^T]$, $\bar{\theta}^h \equiv \theta^T - \bar{\theta}^{cb}$, and $\vartheta_t \equiv \theta_t^{cb} - \bar{\theta}^{cb}$ to capture central bank indexed debt holdings. Then,

$$\theta_t^h = \bar{\theta}^h - \vartheta_t. \quad (38)$$

Clearly, the steady-state central-bank target $\bar{\theta}^{cb}$ co-determines the private sector's steady-state inflation insurance, while the policy variable ϑ_t captures active balance-sheet deviations around that target. The return earned on indexed and non-indexed bonds is the same for each class of holder. What differs across holders is not the cash return, but the valuation of the insurance service provided by indexed debt. Households value indexed bonds because they hedge inflation risk. The central bank receives the same cash return, but does not receive a utility benefit from inflation insurance.

In a first-order approximation of the general equilibrium, any such effects vanish, however. To nonetheless say something about the role of central bank policy, we simply work with a reduced-form wedge to Let ι_{t+1} denote a reduced-form insurance value of holding indexed debt over the period from t to $t + 1$. A positive ι_{t+1} means that households are willing to hold indexed debt at a lower pecuniary yield because the asset provides inflation insurance.

Because reserves and nominal bonds both pay R_t^n , the household's nominal-claim share $(1 - \theta^T) + \theta_t^{cb} = 1 - \theta_t^h$ earns R_t^n , while the indexed share θ_t^h earns $R_t^n \Pi_{t+1}$ in cash plus the insurance service. The household's effective gross return on its portfolio is therefore

$$\mathcal{R}_t^h = R_t^n [1 + \theta_t^h (\Pi_{t+1} (1 + \iota_{t+1}) - 1)]. \quad (39)$$

There is no double-counting of indexation, as θ_t^h is already the privately held indexed share of

household wealth. The purely pecuniary portfolio return, which is what governs household cash flows and (by consolidation) government funding costs, is the same expression with $\iota_{t+1} = 0$.

Log-linearizing around zero inflation and the steady state $(\bar{\theta}^{cb}, \bar{\theta}^h)$ gives

$$\widehat{\mathcal{R}}_t^{h,r} = r_t + \bar{\theta}^h \mathbb{E}_t \pi_{t+1} + \bar{\theta}^h \mathbb{E}_t \iota_{t+1}. \quad (40)$$

Note that the time-varying part of the portfolio share drops out to first order. Since π_{t+1} and ι_{t+1} are zero in steady state, products such as $\vartheta_t \pi_{t+1}$ are second order, so the constant share $\bar{\theta}^h$ suffices. The first term is the standard ex ante real return. The second term is the payoff-indexation channel from the baseline model, but with the privately held indexed share $\bar{\theta}^h$ rather than the issuance share θ^T . The third term is new, capturing the fact that households value indexed debt as inflation insurance. We specify the insurance value using the following linear reduced-form functional form:

$$\iota_{t+1} = \xi_\pi \pi_{t+1} + \xi_\vartheta \vartheta_{t+1}, \quad \xi_\pi, \xi_\vartheta \geq 0. \quad (41)$$

The first term says that the value of inflation insurance is higher when inflation risk is more salient. The second term captures scarcity. When the central bank removes indexed bonds from private portfolios, the remaining privately held indexed bonds provide a more valuable insurance service at the margin.¹⁴

With all these prerequisites ready, we can now define the adjusted household Euler equation, which reads

$$c_t = -\sigma \left(r_t + \bar{\theta}^h \mathbb{E}_t \pi_{t+1} + \bar{\theta}^h \mathbb{E}_t \iota_{t+1} \right) + \mathbb{E}_t c_{t+1}. \quad (42)$$

In deriving the corresponding present-value budget constraint, one distinction relative to the baseline is essential. Recall that the composite coefficient $\Gamma \equiv \beta \left(\sigma\omega - (1 - \beta\omega) \frac{D^{SS}}{Y^{SS}} \right)$ bundles two channels through which returns move consumption, namely, the intertemporal-substitution channel $(\beta\sigma\omega)$ and the wealth-revaluation channel $(-\beta(1 - \beta\omega) \frac{D^{SS}}{Y^{SS}})$, the latter arising because higher realized returns raise household asset income. The pecuniary components of the portfolio return, $r_{t+s} + \bar{\theta}^h \pi_{t+1+s}$, activate both channels and therefore carry the full coefficient Γ . The insurance value ι , in contrast, is non-pecuniary, tilting the consumption path through the Euler equation without generating asset income, so it activates *only* the substitution channel and carries the coefficient $\beta\sigma\omega$. We can therefore express the Permanent Income Hypothesis in this model as

$$\begin{aligned} c_t = & (1 - \beta\omega) \left(a_t + \mathbb{E}_t \sum_{s=0}^{\infty} (\beta\omega)^s (y_{t+s} - t_{t+s}) \right) \\ & - \Gamma \mathbb{E}_t \sum_{s=0}^{\infty} (\beta\omega)^s (r_{t+s} + \bar{\theta}^h \pi_{t+1+s}) - \beta\sigma\omega \bar{\theta}^h \mathbb{E}_t \sum_{s=0}^{\infty} (\beta\omega)^s \iota_{t+1+s}. \end{aligned} \quad (43)$$

¹⁴Since $\theta_t^h = \bar{\theta}^h - \vartheta_t$, an insurance value increasing in ϑ is equivalently one decreasing in the privately held quantity.

Imposing market clearing, $a_t = d_t$ and $c_t = y_t$, using the old fiscal rule

$$t_t = -\varepsilon_t + \tau_d(d_t + \varepsilon_t) + \tau_y y_t, \quad (44)$$

and using the old real-rate rule

$$r_t = \phi y_t, \quad (45)$$

and quasi-differencing (43) gives the recursive aggregate-demand equation

$$\begin{aligned} \mathbb{E}_t y_{t+1} - \frac{\Gamma \bar{\theta}^h}{\beta \omega} \mathbb{E}_t \pi_{t+1} - \sigma \bar{\theta}^h \mathbb{E}_t \iota_{t+1} - (1 - \beta \omega) \mathbb{E}_t d_{t+1} \\ = \frac{\beta \omega (1 - \tau_y) + \Gamma \phi + \tau_y}{\beta \omega} y_t - \frac{(1 - \beta \omega)(1 - \tau_d)}{\beta \omega} (d_t + \varepsilon_t). \end{aligned} \quad (46)$$

Relative to the baseline aggregate-demand equation, the issuance share θ is replaced by the privately held indexed share $\bar{\theta}^h$, and the insurance value ι_{t+1} enters similarly to an additional indexed-return component, but with the factor $\sigma \bar{\theta}^h$, which takes only the perceived benefit from additional insurance. As before, in the representative-agent limit $\omega = 1$, the debt terms cancel out.

We now verify that the government-side accounting is consistent with the household portfolio above. The Treasury pays the return $R_t^n [1 + \theta^T (\Pi_{t+1} - 1)]$ on its entire debt, the central bank earns $R_t^n \Pi_{t+1}$ on its indexed holdings, pays R_t^n on reserves, and remits the difference,

$$\text{Rem}_{t+1} = \theta_t^{cb} (D_t - P_t T_t) R_t^n (\Pi_{t+1} - 1), \quad (47)$$

to the Treasury. Consolidating, the government's liabilities to the private sector are, by the wealth identity (37), equal to D_t , and the pecuniary return the consolidated government pays on them is

$$R_t^n [1 + \theta_t^h (\Pi_{t+1} - 1)], \quad (48)$$

i.e., the nominal rate on the nominal-claim share $1 - \theta_t^h$ (nominal bonds plus reserves) and the indexed cash return on the privately held indexed share θ_t^h . Equivalently, remittance (47) exactly rebates to the Treasury the indexation payments on the share θ_t^{cb} that the central bank holds, reducing the Treasury's net indexation exposure from θ^T to θ_t^h .

The consolidated flow budget constraint is therefore the baseline constraint with θ replaced by θ_t^h . Because the product of the time-varying portfolio deviation ϑ_t and (surprise) inflation is second order around a zero-inflation steady state, the first-order valuation equation depends only on the steady-state share $\bar{\theta}^h = \theta^T - \bar{\theta}^{cb}$:

$$d_{t+1} = \frac{1}{\beta} (d_t - t_t) + \bar{d} r_t - \bar{d} [(1 - \bar{\theta}^h) \pi_{t+1} - \mathbb{E}_t \pi_{t+1}], \quad \bar{d} \equiv \frac{D^{ss}}{Y^{ss}}. \quad (49)$$

Substituting the monetary and fiscal rules and taking expectations gives

$$\mathbb{E}_t d_{t+1} - \bar{d} \bar{\theta}^h \mathbb{E}_t \pi_{t+1} = \left(\bar{d} \phi - \frac{\tau_y}{\beta} \right) y_t + \frac{1 - \tau_d}{\beta} (d_t + \varepsilon_t). \quad (50)$$

To close the policy block of the model, we must specify the evolution of ϑ_t , the central bank's balance sheet instrument by which it can absorb some indexed debt. The central bank in our model retains the real short rate instrument, as given by equation (45). As for the central bank's balance sheet rule, we specify it in deviations from the steady-state target:

$$\vartheta_{t+1} = \rho_\vartheta \vartheta_t + \chi_\pi \mathbb{E}_t \pi_{t+1} + \chi_y y_t + \varepsilon_t^\vartheta, \quad \rho_\vartheta \in [0, 1), \quad \chi_\pi, \chi_y \geq 0. \quad (51)$$

Under this rule, next period's portfolio is chosen at time t on the basis of time- t information: ϑ_{t+1} is predetermined, the rule holds exactly, and the shock ε_t^ϑ is an announced portfolio (QE-type) innovation that takes effect one period after its announcement. This timing mirrors the operational reality that asset-purchase programs are announced before they are implemented, and it will matter below for the count of predetermined variables.

Combining the QE rule with the NKPC ($\pi_t = \kappa y_t + \beta \mathbb{E}_t \pi_{t+1}$), one may also write

$$\vartheta_{t+1} = \left(\chi_y - \frac{\chi_\pi \kappa}{\beta} \right) y_t + \frac{\chi_\pi}{\beta} \pi_t + \rho_\vartheta \vartheta_t + \varepsilon_t^\vartheta. \quad (52)$$

We can now characterize the full model as a system of four equations in four variables. Define, for the following,

$$A_\pi \equiv \frac{\Gamma \bar{\theta}^h}{\beta \omega} + \sigma \bar{\theta}^h \zeta_\pi, \quad A_\vartheta \equiv \sigma \bar{\theta}^h \zeta_\vartheta. \quad (53)$$

Using equation (41), the AD equation can then be written as

$$\mathbb{E}_t y_{t+1} - A_\pi \mathbb{E}_t \pi_{t+1} - A_\vartheta \mathbb{E}_t \vartheta_{t+1} - (1 - \beta \omega) \mathbb{E}_t d_{t+1} = C_y y_t - C_d (d_t + \varepsilon_t), \quad (54)$$

where $C_y \equiv \frac{\beta \omega (1 - \tau_y) + \Gamma \phi + \tau_y}{\beta \omega}$, $C_d \equiv \frac{(1 - \beta \omega)(1 - \tau_d)}{\beta \omega}$. Collecting aggregate demand, the NKPC, the debt valuation equation, and the balance-sheet policy rule gives the following first-order system:

$$\mathbb{E}_t \begin{bmatrix} y_{t+1} \\ \pi_{t+1} \\ d_{t+1} \\ \vartheta_{t+1} \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{bmatrix} \begin{bmatrix} y_t \\ \pi_t \\ d_t + \varepsilon_t \\ \vartheta_t \end{bmatrix} + \begin{bmatrix} A_\vartheta \\ 0 \\ 0 \\ 1 \end{bmatrix} \varepsilon_t^\vartheta, \quad (55)$$

where the entries are:

$$M_{11} = C_y - \frac{\kappa}{\beta} A_\pi + A_\vartheta \left(\chi_y - \frac{\chi_\pi \kappa}{\beta} \right) + (1 - \beta\omega) \left(\bar{d}\phi - \frac{\tau_y}{\beta} - \frac{\bar{d}\bar{\theta}^h \kappa}{\beta} \right), \quad (56)$$

$$M_{12} = \frac{A_\pi}{\beta} + \frac{A_\vartheta \chi_\pi}{\beta} + (1 - \beta\omega) \frac{\bar{d}\bar{\theta}^h}{\beta}, \quad (57)$$

$$M_{13} = \frac{(1 - \beta\omega)(1 - \tau_d)}{\beta} \left(1 - \frac{1}{\omega} \right), \quad (58)$$

$$M_{14} = A_\vartheta \rho_\vartheta, \quad (59)$$

$$M_{21} = -\frac{\kappa}{\beta}, \quad M_{22} = \frac{1}{\beta}, \quad M_{23} = 0, \quad M_{24} = 0, \quad (60)$$

$$M_{31} = \bar{d}\phi - \frac{\tau_y}{\beta} - \frac{\bar{d}\bar{\theta}^h \kappa}{\beta}, \quad (61)$$

$$M_{32} = \frac{\bar{d}\bar{\theta}^h}{\beta}, \quad M_{33} = \frac{1 - \tau_d}{\beta}, \quad M_{34} = 0, \quad (62)$$

$$M_{41} = \chi_y - \frac{\chi_\pi \kappa}{\beta}, \quad M_{42} = \frac{\chi_\pi}{\beta}, \quad M_{43} = 0, \quad M_{44} = \rho_\vartheta. \quad (63)$$

The M_{11} entry differs from the baseline through three terms, namely, the replacement of the issuance share by the privately held indexed share $\bar{\theta}^h$, the household insurance-value term A_π , and the balance-sheet-policy channel $A_\vartheta(\chi_y - \chi_\pi \kappa / \beta)$.

Even in that extended system, we can still say something about the possible saddle path-stability of the system. The model now has two (quasi-)predetermined variables (the real debt stock d_t and the balance-sheet position ϑ_t) and two non-predetermined variables (y_t and π_t). A unique saddle-path stable equilibrium therefore requires exactly two eigenvalues of M outside the unit circle.

Proposition 3 *Suppose $\xi_\vartheta = 0$. Then:*

- (i) *$A_\vartheta = 0$, so the fourth column of M is $(0, 0, 0, \rho_\vartheta)'$ and the characteristic polynomial factors as*

$$\det(M - \lambda I) = (\rho_\vartheta - \lambda) \det(\tilde{M} - \lambda I),$$

where $\tilde{M}$ is the upper-left 3×3 block in (y_t, π_t, d_t) . Since $\rho_\vartheta \in [0, 1)$ lies inside the unit circle and attaches to the predetermined state ϑ_t , determinacy of the four-variable system is equivalent to exactly two eigenvalues of $\tilde{M}$ lying outside the unit circle. In particular, the feedback coefficients (χ_π, χ_y) and the persistence ρ_ϑ of the balance-sheet rule are irrelevant for equilibrium determinacy.

- (ii) *If in addition $\xi_\pi = 0$, then $\tilde{M}$ coincides entry-by-entry with the transition matrix of the baseline three-variable system evaluated at $\theta = \bar{\theta}^h = \theta^T - \bar{\theta}^{cb}$. Propositions 1 and 2 therefore remain valid with θ replaced by $\bar{\theta}^h$. Thus, the determinacy consequences of indexation are governed by the indexed debt left in private hands, not by the amount issued.*

Proof. (i) With $A_\vartheta = 0$, expand $\det(M - \lambda I)$ along the fourth column, whose only nonzero entry

is $M_{44} - \lambda = \rho_\theta - \lambda$. (ii) Setting $\xi_\pi = \xi_\theta = 0$ in (53) gives $A_\pi = \Gamma \bar{\theta}^h / (\beta \omega)$ and $A_\theta = 0$. Comparing (56)–(62) with the baseline system shows that every occurrence of θ in the baseline transition matrix is replaced by $\bar{\theta}^h$. ■

Proposition 3 is a statement about the irrelevance of balance-sheet feedback absent the scarcity channel, and effectively contains an analogue of the irrelevance result of Wallace (1981). How aggressively the central bank moves its indexed-bond portfolio in response to inflation or output has no bearing on equilibrium uniqueness, although this result may be overturned when indexed bonds carry a scarcity-dependent insurance value. What *does* matter is the steady-state portfolio target, and it matters one-for-one as a reduction of the effective indexed share.

We can take this result and apply it to an interesting phenomenon. Let $\xi_\pi = \xi_\theta = 0$. A central bank that permanently holds the entire indexed bond issuance, $\bar{\theta}^{cb} = \theta^T$, therefore restores the determinacy regions of the model without indexed debt ($\theta = 0$). Conversely, the same operation switches off the transmission of active balance-sheet policy altogether, since $A_\theta \propto \bar{\theta}^h = 0$.

Of course, the baseline model can be recovered by setting

$$\bar{\theta}^{cb} = 0, \quad \theta_0 = 0, \quad \varepsilon_t^\theta = 0 \quad \forall t, \quad \xi_\pi = \xi_\theta = 0, \quad \chi_\pi = \chi_y = 0. \quad (64)$$

Then $\bar{\theta}^h = \theta^T$, $\theta_t = 0$ for all t , indexed bonds provide no additional insurance service beyond their pecuniary payoff, the central bank does not use its balance sheet, and the system collapses to the original three-variable model in (y_t, π_t, d_t) .

The effects of the proposed balance sheet policy

We now analyse numerically what this “indexed debt easing” changes in our model economy.

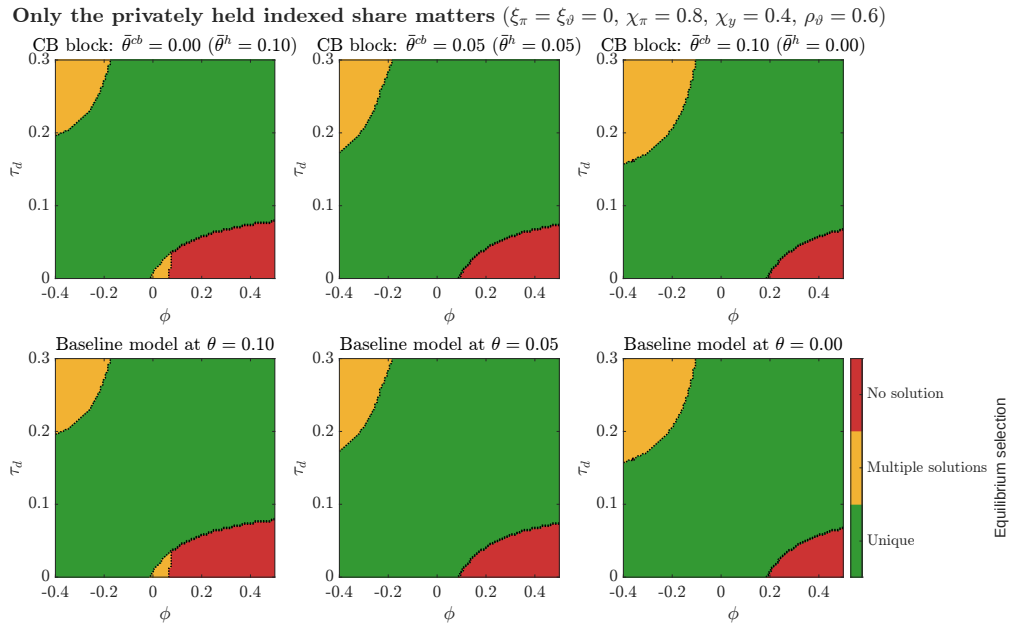


Figure 10: Determinacy properties of the model under the novel monetary instrument (51).

Our first result, which lines up neatly with our discussion of the model under the postulated assumptions, is that mostly only the privately held share of indexed debt matters for the determinacy properties, given the central bank’s ability to swap out such debt in a way that is approximately budget neutral to the consolidated government sector. Figure 10 illustrates this result. The upper row of figure 10 shows the determinacy properties of the model when the central bank swaps indexed debt holdings one-for-one, while the bottom row is the baseline model without the central bank balance sheet instrument. Each column fixes a relative amount of indexed debt in the economy. Clearly, the swap of indexed bonds to reserves in the household portfolio has barely any effect on the admissible equilibrium space. This should not be a surprise given proposition 3, and simply highlights how the stability of the state-space relies on the properties of the transversality condition ensuring government debt sustainability, irrespective of a possible substitution thereof away from household balance sheets.

However, and as illustrated by figure 11, the substitution does not come without consequences.

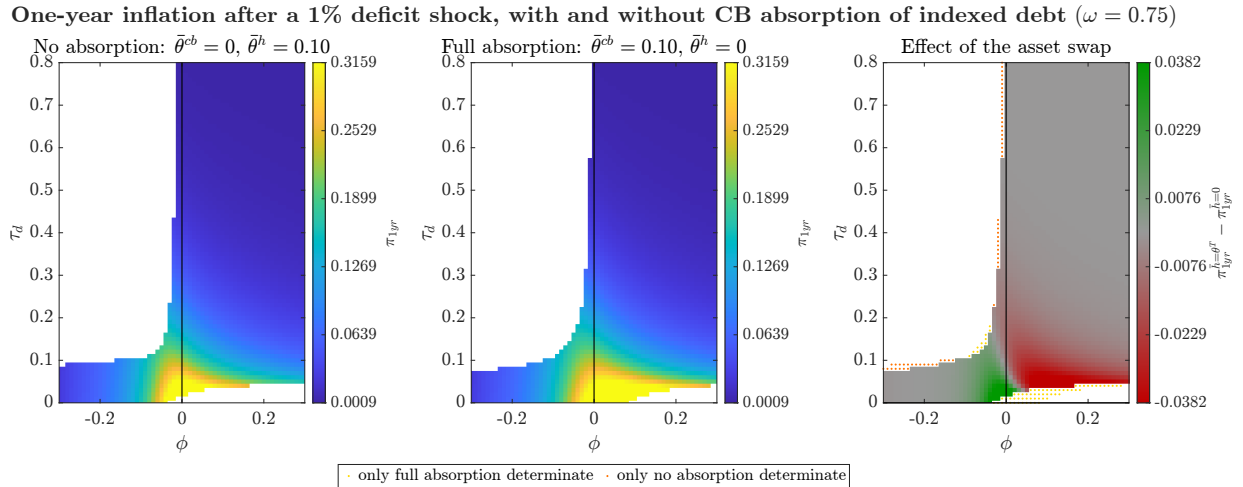


Figure 11: The effect of the asset swap on one-year inflation following a deficit 1% deficit/GDP shock.

The asset swap echoes figure 5 and the discussion of the ‘quasi fiscal dominance’ regime therein: while the deficit shock is inflationary in any case, its effects are principally larger when the fiscal authority actively commits to deficit-finance. In particular, the asset swap has one clear area where it curbs inflation, and one where it exacerbates existing inflationary pressure, with the two areas being approximately akin to the canonical definitions of fiscal and monetary dominance, respectively. The proposed asset swap policy can therefore serve as a supplementary tool for the central bank in fighting fiscally-driven inflation, especially when the central bank does not view it as viable to target the portfolio-adjusted interest rate of households.

We briefly consider the dynamic effects of an announcement by the central bank to engage in such asset swaps. Given our relatively simple modelling framework, the dynamic effects of a surprise innovation in indexed bond buying (through equation (51)) are fairly straightforward, as depicted in equation 12, which shows the effects of such policy by various levels of the postulated insur-

ance value ξ_θ from equation (41).¹⁵ The bond-buying impulse can have a dis-inflationary effect, conditional on the insurance value channel being positive. The bond-buying announcement by the central bank increases the scarcity of indexed bonds for households, who are willing to pay a premium onto their remaining holdings of such bonds. This simultaneously improves the overall fiscal budget balance while limiting the disposable income for households, making this effectively a negative aggregate demand disturbance. Correspondingly, output decreases for some time.

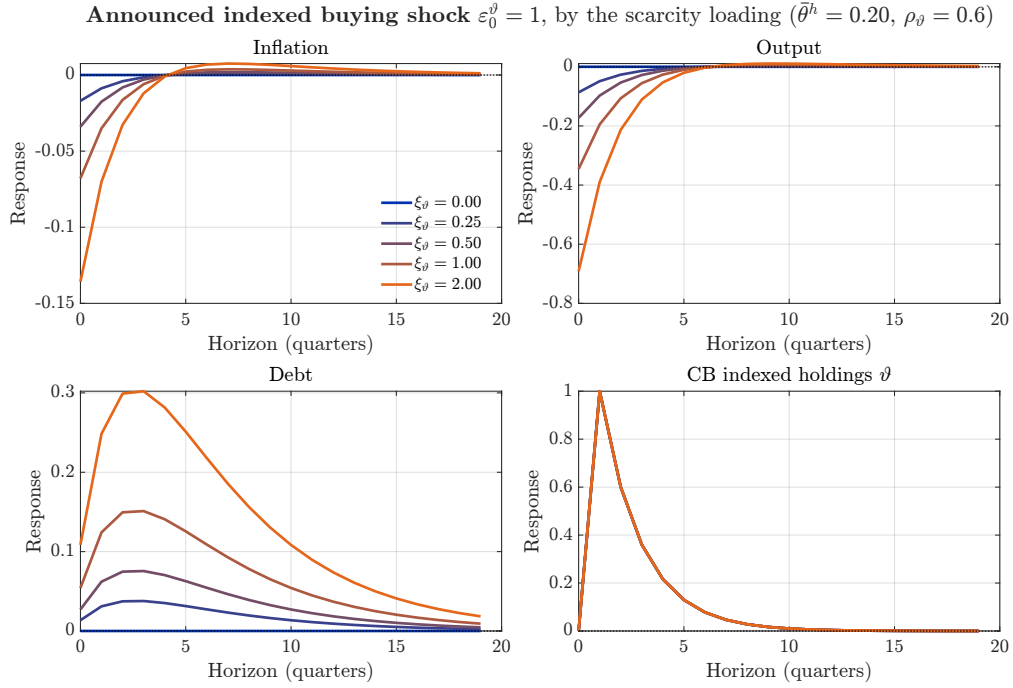


Figure 12: The dynamic effects of an indexed debt-buying shock by the central bank, depending on their insurance value for households.

But even abstracting from the negative effects on output, such activist central bank bond-buying can have unexpected impacts on the economy. In particular, the scarcity channel that affects household demand for indexed bonds through ξ_θ can induce instability for very aggressive central bank bond buying rules, exhibited through high values for the bond buying reaction parameters χ_π and χ_y . Figure 13 illustrates.

The underlying parametrisation is one in which the baseline model exhibits saddle path-stability. Unsurprisingly, when the household does not have any wedge in its valuation of indexed bonds relative to their objective value, the bond-buying policy does not shift the system's stability properties. But once that channel matters ($\xi_\theta > 0$), aggressive bond-buying rules are enabling the existence of sunspot equilibria. Underlying this mechanism is a positive feedback effect between the bond-buying rule and the equilibrium rate of return that the household demands on its portfolio when it values indexed debt in particular. When this complementarity is strong enough, self-fulfilling fluctuations can occur by which central banks end up buying bonds in response to

¹⁵Appendix B.4 provides further insights with varying degrees of persistence of the bond-buying impulse.

inflationary pressure, leading households to demand higher rates of return on their adjusted portfolio, which together end up not violating any equilibrium or market clearing conditions. Such complementarities can lead to increased volatility, despite the intentions of such a policy to curb price level volatility following a fiscal impulse.

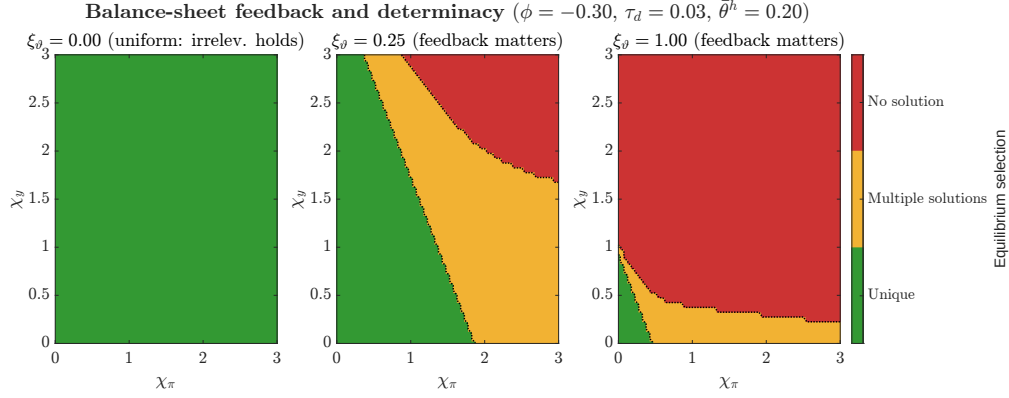


Figure 13: The effects of the balance sheet rule (51) on system stability. The x-axis captures the feedback of inflation on bond-buying through χ_π , while the y-axis varies the feedback of output on bond-buying through χ_y . Each panel is a different value of ξ_θ , the strength of the insurance channel of indexed bonds for households.

5 Conclusion

This paper has looked at fiscal-monetary interactions through the lens of a policy instrument insufficiently considered in dynamic macroeconomic models thus far: inflation-indexed sovereign debt, issued by a multitude of sovereigns.

Inflation-indexed debt highlights two aspects associated with the risks of what is conventionally known as fiscally-led policy mixes. First, recent results emphasizing that the clear distinction between fiscally- and monetary-led policy mixes are overcome once market imperfections such as liquidity risk are added appear to be qualified by the presence of indexed debt: in particular, fiscal policy conventionally considered active is strikingly related to additional inflationary pressure through the presence of inflation-indexed debt, even under such market imperfections. Intuitively, since cost pressures translate into higher debt service burdens for governments when they have issued more indexed debt, monetary and fiscal policies must react sufficiently to cover this additional cost, leading to the emergence of 'quasi fiscally-led' policy mechanisms. Second, it increases the risk of inflationary disasters when we actually are within such fiscally-led policy mixes - that is, when a fiscal deficit shock materializes, it is more inflationary when we actually have higher shares of inflation-indexed debt *and* we are simultaneously operating under policy rules considered to be conventionally fiscally-led.

Other than further robustifying our theoretical results by providing analytic characterizations of medium-run inflationary pressures in the simplified models provided that feature inflation-indexed debt, we believe that the latter part of the paper opened up a lot of questions on the optimal conduct of monetary policy in the face of indexed debt when a multitude of instruments are available to

the monetary authority. There seems to be a scope for quantifying the effects of optimal monetary balance sheet policy, especially relative to the purported ex-ante incentives for a fiscal authority to actually issue indexed debt. Combining these elements, then, yields scope for an analysis of the strategic considerations underlying monetary-fiscal interactions, as alluded to in [Bassetto and Sargent \(2020\)](#). We intend to do so in subsequent research in the near future.

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Appendix

A Appendix for the model in section 2

A.1 Log-linearization of the model in section 2

Log-linearization around the zero-inflation steady-state gives the following system of equations:

1. Nominal bond prices:

$$-\hat{R}_t = (\theta - 1) E_t \hat{\pi}_{t+1}. \quad (\text{A.1})$$

2. Monetary rule:

$$\hat{R}_t = \phi \hat{\pi}_t. \quad (\text{A.2})$$

3. Law of motion of debt:

$$\frac{\pi_t^\theta b_{t-1}}{P_t Y} = \frac{P_t T_t}{P_t Y} + \frac{q_t b_t}{P_t Y} \implies \pi_t^{\theta-1} s_{t-1} R_{t-1} = \tau_t + s_t.$$

Log-linearized:

$$\begin{aligned} (\theta - 1) \hat{\pi}_t + \hat{s}_{t-1} + \hat{R}_{t-1} &= \frac{\tau}{\tau + s} \hat{\tau}_t + \frac{s}{\tau + s} \hat{s}_t. \\ \tau + s = s R \pi^{\theta-1}, \quad \frac{1}{R} = \beta \pi^{\theta-1} &\implies \tau + s = \frac{s}{\beta}, \quad \frac{\tau}{s} = \frac{1 - \beta}{\beta}. \\ (\theta - 1) \hat{\pi}_t + \hat{s}_{t-1} + \hat{R}_{t-1} &= (1 - \beta) \hat{\tau}_t + \beta \hat{s}_t. \end{aligned} \quad (\text{A.3})$$

4. Fiscal rule:

$$\hat{\tau}_t = \gamma \hat{s}_{t-1} + \varphi_t. \quad (\text{A.4})$$

Combining (A.1) and (A.2):

$$\phi \hat{\pi}_t = (1 - \theta) E_t \hat{\pi}_{t+1}, \quad (\text{A.5})$$

Forward (A.3) one period, take expectations at t and use $-\hat{R}_t = (\theta - 1) E_t \hat{\pi}_{t+1}$:

$$\hat{s}_t = (1 - \beta) E_t \hat{\tau}_{t+1} + E_t \beta \hat{s}_{t+1}, \quad (\text{A.6})$$

in which inflation does not directly appear. Substitute in for $\hat{\tau}_{t+1} = \gamma \hat{s}_t + \varphi_{t+1}$ to obtain:

$$\hat{s}_t = (1 - \beta) \gamma \hat{s}_t + (1 - \beta) E_t \varphi_{t+1} + E_t \beta \hat{s}_{t+1}. \quad (\text{A.7})$$

Combining the equations yields the system in section 2.

A.2 The Fisherian model under a standard Taylor Rule

In place of the bond return rate-targeting monetary rule in section 2, we here explore the dynamic properties of the Fisherian framework under a standard policy rule defined on the simple nominal interest rate.

The central bank would then follow a simplified monetary rule on the gross nominal interest rate:

$$R_t^n = \frac{1}{\beta} \Pi_t^\phi, \quad (\text{A.8})$$

where $R_{n,t} \equiv 1 + i_t$ is the gross nominal interest rate, and ϕ is the monetary policy parameter of interest that will indicate whether the economy operates under a monetary-led policy mix or not. This monetary rule implies a zero-inflation steady-state consistent with household optimality (captured by the bond pricing equations), under which $R^n = \frac{1}{\beta}$. Thus, under the present setting, $Q_t = \frac{1}{R_t^n}$, i.e., the price of the nominal bond must be the inverse of the gross nominal interest rate as all bonds are short-term and the relevant interest rate is under control of the monetary authority.

***Equilibrium:** The equilibrium is derived exactly as in the main body of the paper, but making use of the monetary rule (A.8). Expressing the system in log-deviations from a deterministic zero-inflation steady-state, we obtain the following system of equations:

- Nominal bond prices: using $q_t = \frac{1}{R_t}$ and $q_t = \beta \mathbb{E}_t[\Pi_{t+1}^{\theta-1}]$, we get, to a first-order approximation,

$$-\hat{R}_t = (\theta - 1) \mathbb{E}_t \hat{\pi}_{t+1}. \quad (\text{A.9})$$

- Monetary rule: taking logs of the monetary rule, we get

$$\hat{R}_t^n = \phi \hat{\pi}_t. \quad (\text{A.10})$$

- Law of motion of debt: as in the main text,

$$(\theta - 1) \hat{\pi}_t + \hat{s}_{t-1} + \hat{R}_{t-1} = (1 - \beta) \hat{\tau}_t + \beta \hat{s}_t. \quad (\text{A.11})$$

- Fiscal rule: the fiscal rule can be log-linearized exactly:

$$\hat{\tau}_t = \gamma \hat{s}_{t-1} + \varphi_t. \quad (\text{A.12})$$

A no-arbitrage argument: To induce finite demands on either type of asset (and thereby to close the model under the Taylor Rule on the nominal interest rate), which is necessary for the existence of an equilibrium, we posit that households demand equivalent *real* returns from both assets in log-deviations:

$$\hat{R}_t^n - \mathbb{E}_t \hat{\pi}_{t+1} = \hat{R}_t - (1 - \theta) \mathbb{E}_t \hat{\pi}_{t+1}. \quad (\text{A.13})$$

The left-hand side captures the real return on the nominal asset. The right-hand side captures the real return on the government-issued partially-indexed bond, where the inflation adjustment is weighted by $(1 - \theta)$, the *non-indexed share* of that bond.

We can combine equations (A.9)+(A.10)+(A.13) to attain the following system of difference equations:

$$\mathbb{E}_t \hat{\pi}_{t+1} = \phi \hat{\pi}_t, \quad (\text{A.14})$$

For the fiscal block, we remain with the same specification as in the main text:

$$\hat{s}_t = (1 - \beta) \gamma \hat{s}_t + (1 - \beta) \mathbb{E}_t \varphi_{t+1} + \mathbb{E}_t \beta \hat{s}_{t+1}. \quad (\text{A.15})$$

Collecting, the system can be written as

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & \beta \end{pmatrix}}_{\equiv A_0} \mathbb{E}_t \begin{pmatrix} \hat{\pi}_{t+1} \\ \hat{s}_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \phi & 0 \\ 0 & 1 - \gamma(1 - \beta) \end{pmatrix}}_{\equiv A_1} \begin{pmatrix} \hat{\pi}_t \\ \hat{s}_t \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ -(1 - \beta) \end{pmatrix}}_{\equiv C} \mathbb{E}_t \varphi_{t+1}. \quad (\text{A.16})$$

The matrix relevant for the determinacy properties is

$$Z = A_0^{-1} A_1 = \begin{pmatrix} \phi & 0 \\ 0 & \frac{1 - \gamma(1 - \beta)}{\beta} \end{pmatrix}. \quad (\text{A.17})$$

Its eigenvalues are ϕ and $\frac{1 - \gamma(1 - \beta)}{\beta}$. A *monetary-led* equilibrium requires $\phi > 1$ and $\gamma > 1$. A *fiscally-led* equilibrium requires $\phi < 1$ and $\gamma < 1$. In sum, inflation-indexed debt does *not* matter for the determinacy properties of this simple model when the Taylor Rule ignores the composition of government debt. This is a direct consequence of the no-arbitrage argument that forces the real return rate of the partially-indexed government bond to be equal to the return rate of the zero-supply outside asset.

B Appendix for the model in section 3

B.1 An argument ensuring exactly one root inside the unit circle

Proof. We here present a brief argument by which we can derive the conditions under which exactly one root of a 2×2 system lies inside the unit circle.

Recall that the characteristic equation of Z is given by

$$p(\lambda) = \lambda^2 - \text{tr}(Z)\lambda + \det(Z).$$

Define the two following expressions for convenience:

$$p(1) \equiv 1 - \text{tr}(Z) + \det(Z), \quad p(-1) \equiv 1 + \text{tr}(Z) + \det(Z).$$

Now, let the two eigenvalues, as defined by the characteristic equation, be given by λ_1 and λ_2 . Then, $p(\cdot)$ evaluated at 1 and -1 yields:

$$p(1) = (1 - \lambda_1)(1 - \lambda_2), \quad p(-1) = (1 + \lambda_1)(1 + \lambda_2),$$

while the determinant and trace can be expressed as $\det(Z) = \lambda_1\lambda_2$, $\text{tr}(Z) = \lambda_1 + \lambda_2$.

Now, combining the insight of [Woodford \(2011\)](#) on the location of both eigenvalues outside the unit circle with results that allow us to express cases when both eigenvalues are inside the unit circle (e.g., [Antsaklis and Michel \(2006\)](#), Theorem 10.10), we can summarize the necessary and sufficient conditions for the existence of both eigenvalues either inside or outside the unit circle as follows:

- *Both inside* ($|\lambda_1|, |\lambda_2| < 1$) iff

$$\det(Z) < 1, \quad p(1) > 0, \quad p(-1) > 0.$$

- *Both outside (Case I)* iff

$$\det(Z) > 1, \quad p(1) > 0, \quad p(-1) > 0.$$

- *Both outside (Case II)* iff

$$p(1) < 0, \quad p(-1) < 0.$$

In addition, we now use a technical boundary to rule out that $p(-1) < 0$. That technical boundary is expressed as the condition under which $p(-1) > 0$, which is given by:

$$\begin{aligned} & p(-1) > 0 \\ \Leftrightarrow & 2 \left(1 + \frac{1}{\beta} \right) + \phi\sigma \left(1 + \frac{1}{\beta} \right) - \frac{\kappa\sigma\theta}{\beta} > 0 \\ \Leftrightarrow & (2 + \phi\sigma) \left(1 + \frac{1}{\beta} \right) > \frac{\kappa\sigma\theta}{\beta} \\ \Leftrightarrow & \phi > -\frac{1}{\sigma} \left[2 - \frac{\kappa\sigma\theta}{1 + \beta} \right]. \end{aligned}$$

Without indexed debt, this condition is always satisfied under the technical boundary ensuring a positive determinant, $\phi > -\frac{1}{\sigma}$. Here, for high levels of inflation-indexed debt (and a sufficiently high debt-to-GDP ratio or a sufficiently steep Phillips Curve), this might be an additional restriction to consider.

In sum total, we can express a total of eight cases in dependence of $\det(Z)$, $p(1)$, and $p(-1)$, for which six cases are covered so far as necessary and sufficient:

- $\det(Z) < 1, p(1) > 0, p(-1) > 0$: Both roots inside
- $\det(Z) < 1, p(1) > 0, p(-1) < 0$: Ruled out
- $\det(Z) < 1, p(1) < 0, p(-1) > 0$:
- $\det(Z) < 1, p(1) < 0, p(-1) < 0$: Both roots outside (case II)
- $\det(Z) > 1, p(1) > 0, p(-1) > 0$: Both roots outside (case I)
- $\det(Z) > 1, p(1) > 0, p(-1) < 0$: Ruled out
- $\det(Z) > 1, p(1) < 0, p(-1) > 0$:
- $\det(Z) > 1, p(1) < 0, p(-1) < 0$: Both roots outside (case II)

Then, all that must be proven is the restriction under which $p(1) = 1 - \text{tr}(Z) + \det(Z) = \left(\frac{1}{\beta} - 1\right)\phi\sigma + \frac{\kappa\sigma\theta}{\beta}$ is smaller than zero. These are the remaining cases for which exactly one real root lies inside the unit circle. This restriction is given by:

$$\phi < -\frac{\kappa\theta}{1 - \beta} \quad (\text{B.1})$$

Therefore, the region ensuring a saddle-path stable equilibrium under a fiscally-led policy mix restricts the monetary policy parameter ϕ into the following space:

$$\max\left\{-\frac{1}{\sigma}, -\frac{1}{\sigma}\left[2 - \frac{\kappa\sigma\theta}{1 + \beta}\right]\right\} < \phi < -\frac{\kappa\theta}{1 - \beta}. \quad (\text{B.2})$$

For $\theta = 0$, this reverts to the standard restriction for a fiscally-led policy mix in RANK models that $-\frac{1}{\sigma} < \phi < 0$. ■

B.2 Criteria for monetary dominance in the RANK version of the model in section 3

Define again the matrix in equation (24) as A . Its characteristic polynomial is given by:

$$\begin{aligned} \det(A - \lambda I) &= \left[\frac{1}{\beta}(1 - \tau_d) - \lambda\right] \left[\left(1 + \phi\sigma - \frac{\kappa\sigma\theta}{\beta} - \lambda\right)\left(\frac{1}{\beta} - \lambda\right) + \frac{\kappa\sigma\theta}{\beta^2}\right] \\ &= \left[\frac{1}{\beta}(1 - \tau_d) - \lambda\right] \left[\lambda^2 - \lambda\left(\frac{1}{\beta} + 1 + \sigma\phi - \frac{\kappa\sigma\theta}{\beta}\right) + \frac{1}{\beta}(1 + \sigma\phi)\right] \end{aligned} \quad (\text{B.1})$$

In order to establish how indexed debt influences the system's stability properties, we consider the

boundaries under which conventional monetary dominance is established.

Monetary dominance requires fiscal passivity, which here is ensured by $\tau_d > 1 - \beta$, i.e., fiscal policy must be sufficiently reactive to deviations of the value of government debt from its steady-state. In that case, the root implied by the first bracket in equation (B.1) is inside the unit circle, so we require two further roots outside the unit circle.

Analysing those can be done equivalently by reducing our focus to the 2×2 sub-system on (y_t, π_t) , characterized by the following matrix Z :

$$Z \equiv \begin{bmatrix} 1 + \phi\sigma - \frac{\kappa\sigma\theta}{\beta} & \frac{\sigma\theta}{\beta} \\ -\frac{\kappa}{\beta} & \frac{1}{\beta} \end{bmatrix}.$$

Following [Woodford \(2011\)](#) and [Rachel and Ravn \(2025\)](#), either of the following sets of conditions is sufficient to guarantee that two roots of Z lie outside the unit circle:

1. First case:

- $\det(Z) > 1$
- $\det(Z) - \text{tr}(Z) > -1$
- $\det(Z) + \text{tr}(Z) > -1$

2. Second case:

- $\det(Z) - \text{tr}(Z) < -1$
- $\det(Z) + \text{tr}(Z) < -1$.

We focus on case 1, ruling out case 2 along the way. First, note that:

$$\begin{aligned} \det(Z) &= \frac{1}{\beta}(1 + \sigma\phi) \\ \text{tr}(Z) &= (1 + \sigma\phi) + \frac{1}{\beta}[1 - \kappa\sigma\theta]. \end{aligned}$$

Let active monetary policy be characterized by $\phi \geq 0$, as in the main body of the text. Then, $\det(Z) > 1$ is trivially true for any $\beta < 1$. Next, we consider the second condition of the first case:

$$\begin{aligned} \det(Z) - \text{tr}(Z) &= \frac{1}{\beta}(\sigma\phi + \kappa\sigma\theta) - (1 + \sigma\phi) > -1 \\ \Leftrightarrow \left(\frac{1}{\beta} - 1\right)\sigma\phi + \frac{\kappa\sigma\theta}{\beta} &> 0, \end{aligned}$$

which, again, holds trivially true for all values of θ , including $\theta = 0$ (which is the base case of [Angeletos et al. \(2024\)](#) and [Rachel and Ravn \(2025\)](#)). Therefore, the second set of conditions must not be considered, since the previous equation implies that $\det(Z) - \text{tr}(Z) < -1$ cannot hold true for $\phi > 0$.

Finally, consider the third condition of the first case. Here, we find that:

$$\begin{aligned}
\det(Z) + \text{tr}(Z) &= \left(\frac{1}{\beta} + 1\right)(1 + \sigma\phi) + \frac{1}{\beta}(1 - \kappa\sigma\theta) > -1 \\
&\Leftrightarrow (1 + \beta)(1 + \sigma\phi) + (1 - \kappa\sigma\theta) > -\beta \\
&\Leftrightarrow (1 + \beta)(1 + \sigma\phi) > \kappa\sigma\theta - (1 + \beta) \\
&\Leftrightarrow 1 + \sigma\phi > \frac{\kappa\sigma\theta}{1 + \beta} - 1 \\
&\Leftrightarrow \phi > \frac{1}{\sigma} \left[\frac{\kappa\sigma\theta}{1 + \beta} - 2 \right],
\end{aligned}$$

i.e., the admissible bounds for ϕ that ensure monetary dominance *can* tighten when the share of indexed debt is sufficiently high. Tighter determinacy bounds under monetary dominance due to the presence of indexed debt are supported through steep Phillips Curves (high κ) or a high elasticity of intertemporal substitution (high σ), equivalently a low coefficient of relative risk aversion ($1/\sigma$). Note that without indexed debt ($\theta = 0$), this condition reduces to $\phi > -\frac{2}{\sigma}$, which is trivially true.

B.3 Proof of the characterization of the parametric boundaries ensuring saddle path-stable equilibria

Here, we first provide a root-count characterization before translating it into the parametric boundaries used in the main text. Define, as in the main text,

$$\mathcal{J} \equiv (1 - \beta\omega)\left(1 - \frac{1}{\omega}\right), \quad \mathcal{S} \equiv \sigma + \frac{D^{SS}}{Y^{SS}} \mathcal{J}.$$

Under mortality risk, $\omega \in (0, 1)$ implies $\mathcal{J} < 0$. We impose the following assumptions throughout.

Assumption 1 *The elasticity-of-intertemporal-substitution component dominates the debt-weighted mortality wedge:*

$$\mathcal{S} > 0.$$

Assumption 2 *The degree of inflation indexation satisfies*

$$\theta < \bar{\theta} \equiv \frac{1 + \beta}{\kappa\sigma}.$$

We also maintain $\tau_y > 0$, $\tau_d \in [0, 1]$, and $\phi > -1/\sigma$. Multiplying the inverse of the left-hand

matrix in the main text into its right-hand matrix gives

$$A = \begin{bmatrix} 1 + \phi\varsigma - \frac{\kappa\theta}{\beta}\varsigma - \frac{\tau_y}{\beta}\mathcal{T} & \frac{\theta}{\beta}\varsigma & \frac{\mathcal{T}}{\beta}(1 - \tau_d) \\ -\frac{\kappa}{\beta} & \frac{1}{\beta} & 0 \\ \frac{D^{SS}}{Y^{SS}}\phi - \frac{\tau_y}{\beta} - \frac{\kappa\theta}{\beta}\frac{D^{SS}}{Y^{SS}} & \frac{\theta}{\beta}\frac{D^{SS}}{Y^{SS}} & \frac{1-\tau_d}{\beta} \end{bmatrix}.$$

Its trace is

$$\text{tr}(A) = 1 + \phi\varsigma - \frac{\kappa\theta}{\beta}\varsigma - \frac{\tau_y}{\beta}\mathcal{T} + \frac{2 - \tau_d}{\beta}. \quad (\text{B.1})$$

Let $S_2(A)$ denote the sum of the principal 2×2 minors. This can be expressed as:

$$S_2(A) = \frac{1}{\beta^2} \left[\beta \frac{D^{SS}}{Y^{SS}} \mathcal{T} \phi - \mathcal{T} \tau_y - \beta \phi \sigma \tau_d + 2\beta \phi \sigma - \beta \tau_d + 2\beta + \kappa \sigma \theta (\tau_d - 1) - \tau_d + 1 \right]. \quad (\text{B.2})$$

The determinant simplifies to

$$\det(A) = \frac{1 - \tau_d}{\beta^2} (1 + \sigma \phi). \quad (\text{B.3})$$

This simplification follows because the θ -, τ_y -, and debt-ratio terms cancel once $\varsigma - \frac{D^{SS}}{Y^{SS}} \mathcal{T} = \sigma$ is used.

Write the characteristic polynomial as

$$p(\lambda) \equiv \det(\lambda I - A) = \lambda^3 + \Gamma_2 \lambda^2 + \Gamma_1 \lambda + \Gamma_0, \quad (\text{B.4})$$

where

$$\Gamma_2 = -\text{tr}(A), \quad \Gamma_1 = S_2(A), \quad \Gamma_0 = -\det(A). \quad (\text{B.5})$$

Equation (B.3) and the maintained parameter restrictions imply $\Gamma_0 \leq 0$.

Now, let $p(\lambda) = \lambda^3 + \Gamma_2 \lambda^2 + \Gamma_1 \lambda + \Gamma_0$ have real coefficients and no roots on the unit circle. Suppose that

$$\Gamma_0 \leq 0, \quad p(-1) < 0, \quad p(1) > 0.$$

Following [Woodford \(2011\)](#) and [Rachel and Ravn \(2025\)](#), then, exactly two roots of p lie outside the unit circle if and only if either

$$\Gamma_0(\Gamma_0 - \Gamma_2) + \Gamma_1 - 1 > 0, \quad (\text{B.6})$$

or

$$\Gamma_0(\Gamma_0 - \Gamma_2) + \Gamma_1 - 1 < 0 \quad \text{and} \quad |\Gamma_2| > 3. \quad (\text{B.7})$$

To populate the equations, we now establish the signs of $p(-1)$ and $p(1)$ in the present model.

The sign at $\lambda = -1$. Since

$$p(-1) = -\det(I + A),$$

it suffices to establish $\det(I + A) > 0$. Define

$$\Sigma(\tau_d) \equiv (1 + \beta - \tau_d)\sigma + \beta \frac{D^{SS}}{Y^{SS}} \mathcal{J}.$$

Because $\delta > 0$ and $\tau_d \leq 1$,

$$\Sigma(\tau_d) \geq \beta\delta > 0.$$

Direct expansion gives

$$\beta^2 \det(I + A) = (1 + \beta) \left[2(1 + \beta - \tau_d) - \tau_y \mathcal{J} \right] + \Sigma(\tau_d) [(1 + \beta)\phi - \kappa\theta]. \quad (\text{B.8})$$

The right-hand side is strictly increasing in ϕ . Its infimum over $\phi > -1/\sigma$ is therefore obtained as $\phi \searrow -1/\sigma$. At that boundary,

$$\beta^2 \det(I + A) = (1 + \beta)R(\tau_d) - \kappa\theta\Sigma(\tau_d),$$

where

$$R(\tau_d) \equiv (1 + \beta - \tau_d) + |\mathcal{J}| \left(\tau_y + \frac{\beta}{\sigma} \frac{D^{SS}}{Y^{SS}} \right) > 0.$$

The ratio $R(\tau_d)/\Sigma(\tau_d)$ is non-decreasing in τ_d , and at $\tau_d = 0$,

$$R(0) \geq 1 + \beta, \quad \Sigma(0) \leq (1 + \beta)\sigma.$$

It follows that

$$\frac{(1 + \beta)R(\tau_d)}{\Sigma(\tau_d)} \geq \frac{1 + \beta}{\sigma}.$$

The bound $\theta < (1 + \beta)/(\kappa\sigma)$ therefore guarantees

$$\det(I + A) > 0, \quad p(-1) < 0. \quad (\text{B.9})$$

The sign at $\lambda = 1$. Define

$$C(\tau_d) \equiv \sigma(\tau_d - (1 - \beta)) + \beta \mathcal{J} \frac{D^{SS}}{Y^{SS}}.$$

Using this in $p(1)$ then gives

$$p(1) = \det(I - A) = \frac{1}{\beta^2} \left\{ C(\tau_d) [\theta\kappa + (1 - \beta)\phi] - \tau_y \mathcal{J} (1 - \beta) \right\}. \quad (\text{B.10})$$

Let

$$\tau_d^0 \equiv 1 - \beta - \frac{\beta \mathcal{J}}{\sigma} \frac{D^{SS}}{\gamma^{SS}}.$$

Then $C(\tau_d^0) = 0$, $C(\tau_d) < 0$ for $\tau_d < \tau_d^0$, and $C(\tau_d) > 0$ for $\tau_d > \tau_d^0$. When $C(\tau_d) \neq 0$, define

$$\phi^*(\tau_d) \equiv \frac{\tau_y \mathcal{J}}{C(\tau_d)} - \frac{\theta \kappa}{1 - \beta}. \quad (\text{B.11})$$

Equation (B.10) therefore implies

$$p(1) > 0 \iff \begin{cases} \phi < \phi^*(\tau_d), & \tau_d < \tau_d^0, \\ \phi > \phi^*(\tau_d), & \tau_d > \tau_d^0. \end{cases} \quad (\text{B.12})$$

At $\tau_d = \tau_d^0$,

$$p(1) = -\frac{\tau_y \mathcal{J} (1 - \beta)}{\beta^2} > 0.$$

Furthermore,

$$\frac{\partial \phi^*(\tau_d)}{\partial \tau_d} = -\frac{\tau_y \mathcal{J} \sigma}{C(\tau_d)^2} > 0, \quad \frac{\partial \phi^*(\tau_d)}{\partial \theta} = -\frac{\kappa}{1 - \beta} < 0.$$

The remaining root-count condition. Define

$$\Delta(\phi, \tau_d) \equiv 1 - S_2(A) + \text{tr}(A) \det(A) - \det(A)^2. \quad (\text{B.13})$$

Using $\Gamma_2 = -\text{tr}(A)$, $\Gamma_1 = S_2(A)$, and $\Gamma_0 = -\det(A)$ gives

$$\Gamma_0(\Gamma_0 - \Gamma_2) + \Gamma_1 - 1 = -\Delta(\phi, \tau_d).$$

Combining equations (B.6)-(B.7) with equations (B.9) and (B.10) proves that exactly two roots of A lie outside the unit circle if and only if

$$p(1) > 0 \quad (\text{B.14})$$

and either

$$\Delta(\phi, \tau_d) < 0, \quad (\text{B.15})$$

or

$$\Delta(\phi, \tau_d) > 0 \quad \text{and} \quad |\text{tr}(A)| > 3. \quad (\text{B.16})$$

This is the exact determinacy characterization stated in Proposition 2.

Parametric bands. For every fixed τ_d , Δ is a quadratic polynomial in ϕ :

$$\Delta(\phi, \tau_d) = \frac{1}{\beta^4} [\alpha(\tau_d)\phi^2 + b(\tau_d; \theta)\phi + c(\tau_d; \theta)], \quad (\text{B.17})$$

where the coefficients are unambiguously defined from equation (B.13) by

$$\alpha(\tau_d) \equiv \frac{\beta^4}{2} \frac{\partial^2 \Delta}{\partial \phi^2}, \quad b(\tau_d; \theta) \equiv \beta^4 \frac{\partial \Delta}{\partial \phi} \Big|_{\phi=0}, \quad c(\tau_d; \theta) \equiv \beta^4 \Delta(0, \tau_d).$$

The leading coefficient simplifies to

$$\alpha(\tau_d) = \sigma(1 - \tau_d) \left[\beta^2 \frac{D^{SS}}{Y^{SS}} \mathcal{I} + \sigma(\beta^2 - (1 - \tau_d)) \right]. \quad (\text{B.18})$$

Define

$$\tau_d^\# \equiv 1 - \left[\beta^2 + \frac{\beta^2}{\sigma} \frac{D^{SS}}{Y^{SS}} \mathcal{I} \right]. \quad (\text{B.19})$$

Then $\alpha(\tau_d) > 0$ for $\tau_d^\# < \tau_d < 1$. Let

$$\mathcal{D}(\tau_d; \theta) \equiv b(\tau_d; \theta)^2 - 4\alpha(\tau_d)c(\tau_d; \theta).$$

Whenever $\tau_d^\# < \tau_d < 1$ and $\mathcal{D}(\tau_d; \theta) > 0$, define

$$\phi_\pm(\tau_d; \theta) \equiv \frac{-b(\tau_d; \theta) \pm \sqrt{\mathcal{D}(\tau_d; \theta)}}{2\alpha(\tau_d)}. \quad (\text{B.20})$$

Convexity then implies

$$\Delta(\phi, \tau_d) < 0 \iff \phi_-(\tau_d; \theta) < \phi < \phi_+(\tau_d; \theta). \quad (\text{B.21})$$

This band must be intersected with the $p(1) > 0$ region in equation (B.12), it is not by itself sufficient for determinacy.

The trace can be written as

$$\text{tr}(A) = C_0(\tau_d; \theta) + \mathcal{S}\phi,$$

where

$$C_0(\tau_d; \theta) \equiv 1 + \frac{2 - \tau_d}{\beta} - \frac{\kappa\theta}{\beta} \mathcal{S} - \frac{\tau_y}{\beta} \mathcal{I}.$$

Define

$$\phi_{\text{tr}}^\pm(\tau_d; \theta) \equiv \frac{\pm 3 - C_0(\tau_d; \theta)}{\mathcal{S}}. \quad (\text{B.22})$$

Since $\mathcal{S} > 0$,

$$|\text{tr}(A)| > 3 \iff \phi < \phi_{\text{tr}}^-(\tau_d; \theta) \quad \text{or} \quad \phi > \phi_{\text{tr}}^+(\tau_d; \theta).$$

Thus, when $\alpha > 0$ and $\mathcal{D} > 0$, the full determinate set is the intersection of equation (B.12) with

$$\begin{aligned} & \{\phi_-(\tau_d; \theta) < \phi < \phi_+(\tau_d; \theta)\} \\ & \cup (\{\phi < \phi_-(\tau_d; \theta) \text{ or } \phi > \phi_+(\tau_d; \theta)\} \cap \{\phi < \phi_{\text{tr}}^-(\tau_d; \theta) \text{ or } \phi > \phi_{\text{tr}}^+(\tau_d; \theta)\}). \end{aligned} \quad (\text{B.23})$$

Equation (B.23) displays both the central band and the possible outer wedge. For values of τ_d at which $\alpha \leq 0$ or $\mathcal{D} \leq 0$, the exact conditions (B.10), (B.15), and (B.16) continue to apply even though the boundary is not represented by two distinct convex-band roots.

Comparative statics in θ . The exact derivative of Δ with respect to indexation is

$$\frac{\partial \Delta}{\partial \theta} = \frac{\kappa(1 - \tau_d)}{\beta^3} [\beta\sigma - \mathcal{J}(1 + \sigma\phi)]. \quad (\text{B.24})$$

Define

$$\phi_\theta^* \equiv \frac{\beta}{\mathcal{J}} - \frac{1}{\sigma}.$$

For a fixed policy pair (ϕ, τ_d) , increasing θ raises Δ when $\phi < \phi_\theta^*$ and lowers Δ when $\phi > \phi_\theta^*$. Consequently, increasing indexation locally tightens the condition $\Delta < 0$ below ϕ_θ^* and loosens it above ϕ_θ^* .

For a regular boundary root $\phi_j(\tau_d; \theta)$, $j \in \{-, +\}$, implicit differentiation gives

$$\frac{\partial \phi_j}{\partial \theta} = - \frac{\partial \Delta / \partial \theta}{\partial \Delta / \partial \phi} \Big|_{\phi=\phi_j}. \quad (\text{B.25})$$

Under convexity, $\partial \Delta / \partial \phi < 0$ at ϕ_- and $\partial \Delta / \partial \phi > 0$ at ϕ_+ . Equation (B.25), rather than the sign of equation (B.24) alone, determines how each edge of the band moves.

Finally,

$$\frac{\partial \phi^*}{\partial \theta} = - \frac{\kappa}{1 - \beta} < 0, \quad \frac{\partial \phi_{\text{tr}}^\pm}{\partial \theta} = \frac{\kappa}{\beta} > 0.$$

Indexation therefore shifts the $p(1)$ boundary to the left and the trace boundaries to the right, while its effect on the Δ -band depends on the location of the policy pair. There is no globally monotone statement that higher θ must shrink the determinate policy set. This completes the proof.

B.4 Additional numerical results

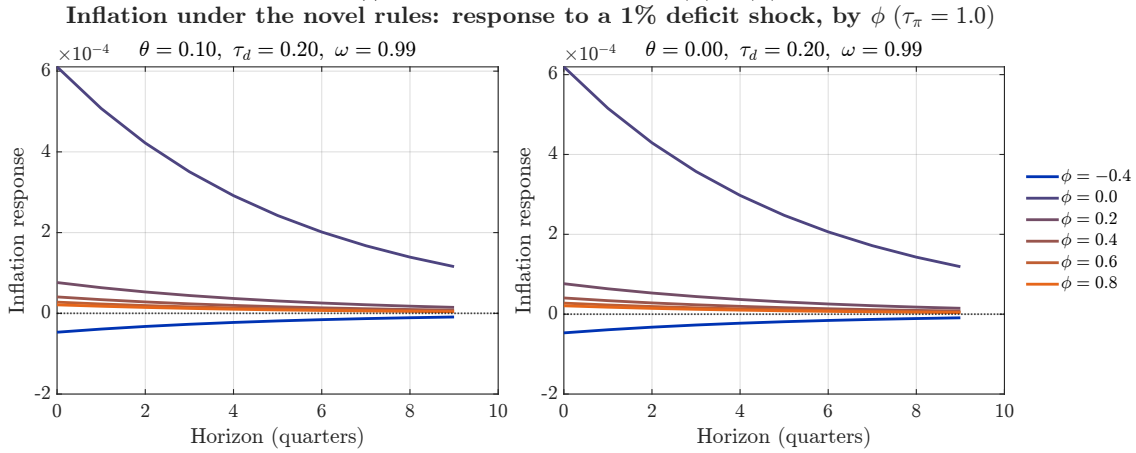
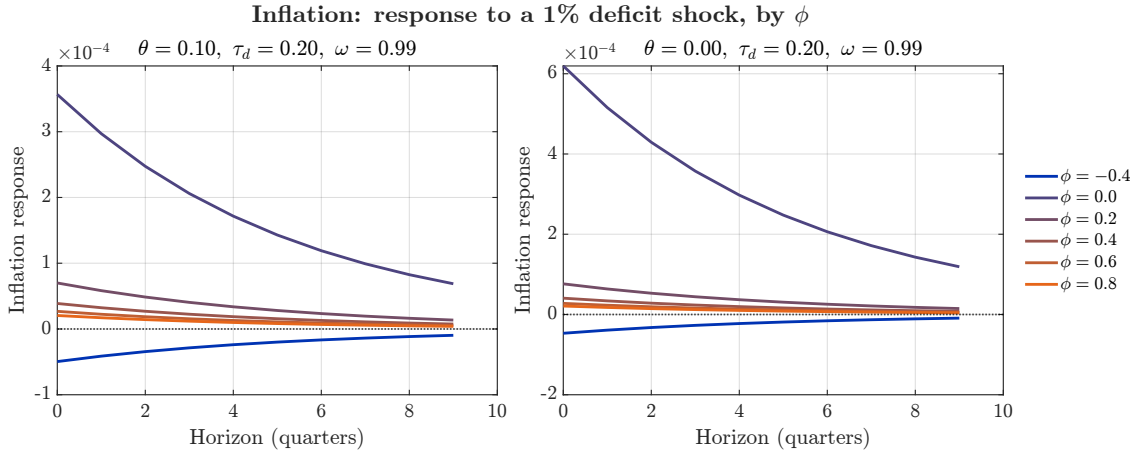
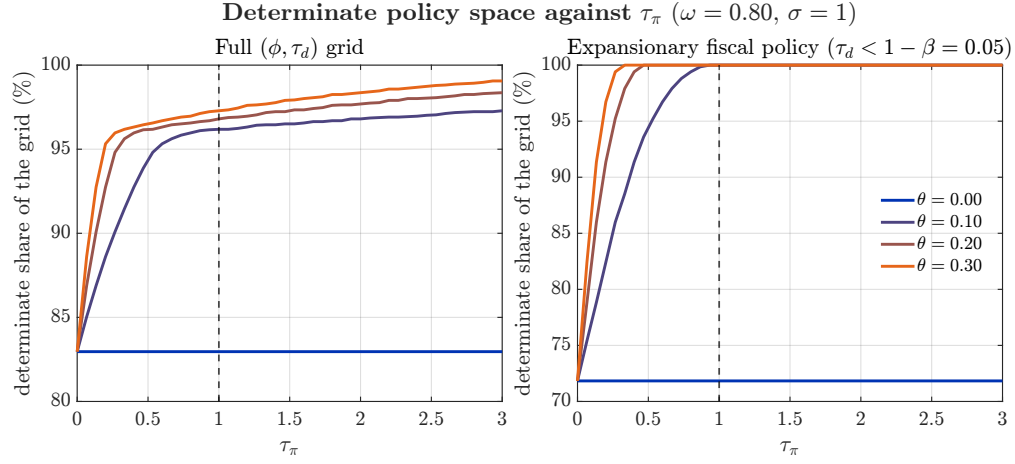


Figure B.2: A comparison of the inflationary effects of a 1% deficit/GDP shock under the baseline and alternative policy rules.

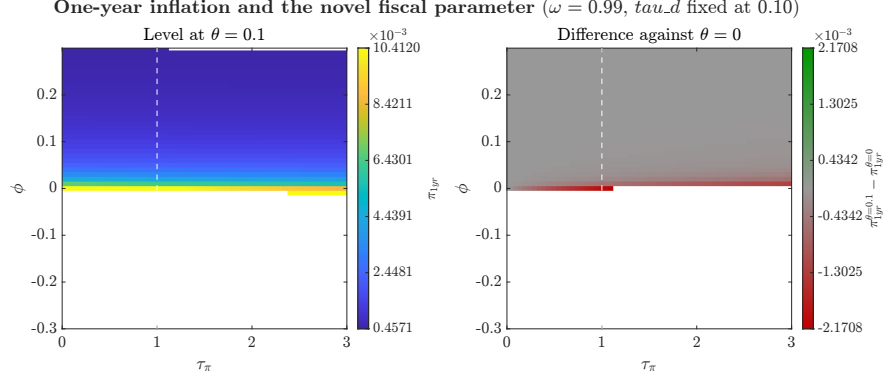


Figure B.3: The effects of a fiscal impulse on the price level under the novel monetary rule (31) and the novel fiscal rule (32), varying the main monetary reaction parameter on the y-axis and τ_π on the x-axis.

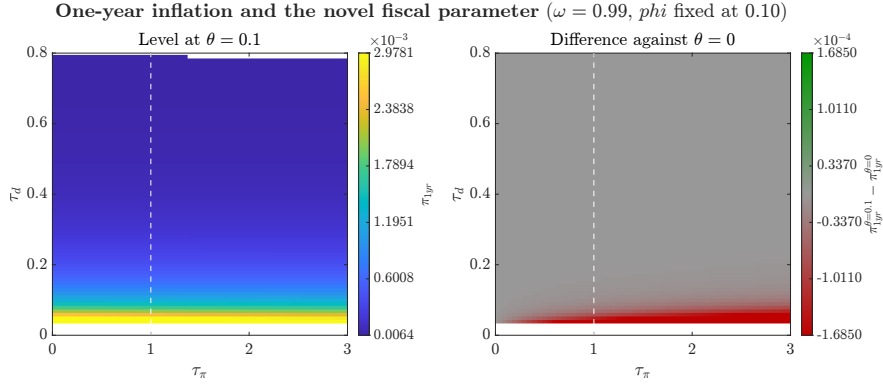


Figure B.4: The effects of a fiscal impulse on the price level under the novel monetary rule (31) and the novel fiscal rule (32), varying the main fiscal reaction parameter on the y-axis and τ_π on the x-axis.

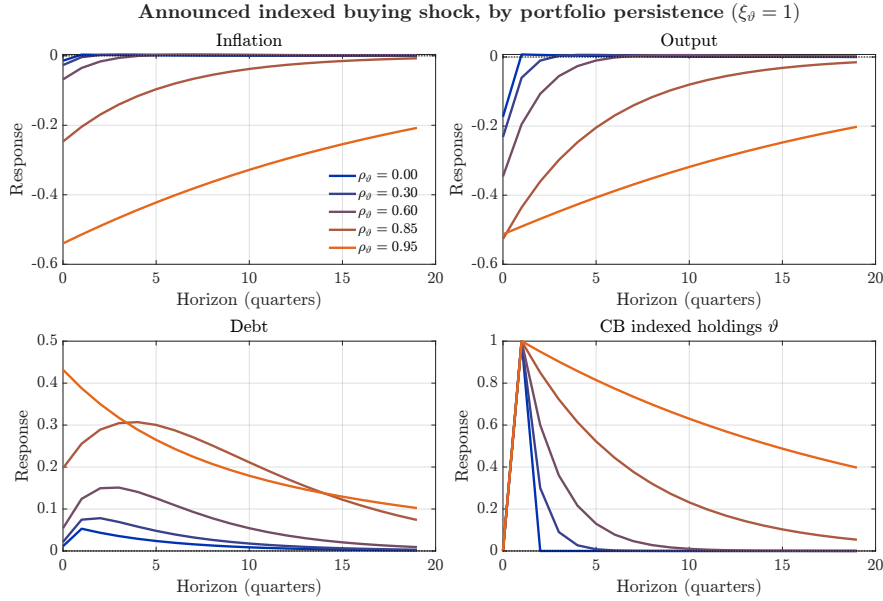


Figure B.5: Effects of the central bank asset swap through equation (51) by various degrees of persistence.

C Fiscally-led Policy Mixes and Materialized Inflation empirically

We now supplement the theoretical results with an empirical analysis evaluating the possibility of fiscally-led policy regimes. Given that empirically distinguishing between two such clear-cut regimes is a futile task (Neumeyer and Nicolini, 2025), we mostly provide correlational evidence that hints at the relevance of our indexed debt mechanism in periods that are more likely to be associated with fiscally-led policy mixes. At the end of this appendix, we utilize exogenously identified fiscal shocks and an exogenously identified regime classification to provide quasi-causal evidence in favour of our mechanism for the specific context of the US.

C.1 Correlations and non-causal regressions: the coincidence of probable fiscally-led policy mixes and inflation-indexed debt

At first, we intend to provide certain aspects motivating our analysis of inflation-indexed debt as a potential driver of inflationary episodes, as well as of the broad relevance of our indexed debt mechanism.

To do so, we first examine in a simple descriptive statistic whether inflation-indexed debt levels are descriptively related to materialized inflation rates. This task is conducted in figure C.6, which plots the share of inflation-indexed debt against observed gross annual CPI inflation rates in the sample of BIS Member Countries issuing inflation-indexed debt since 1990.

While only correlational, the evidence presented here hinders the traditional message of inflation-indexed debt serving as a device ensuring that issuing governments will successfully curb inflationary pressure (Campbell and Shiller, 1996): there seems to be a clear positive correlation between observed shares of indexed debt and annual inflation rates in our long-running sample.

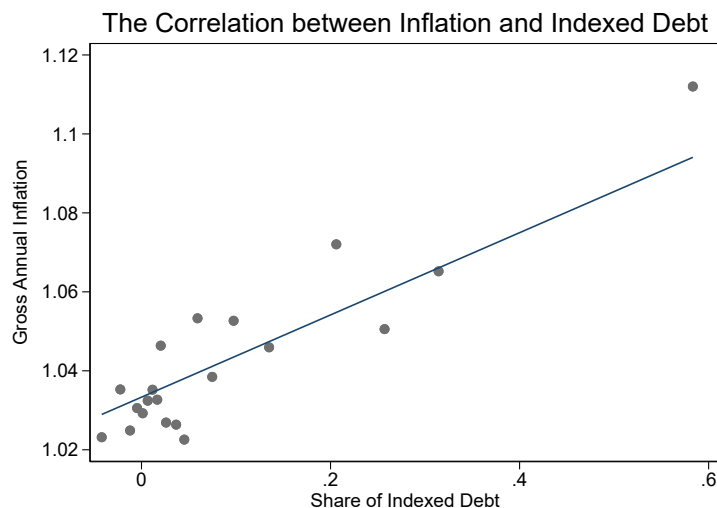


Figure C.6: Binscatter of inflation-indexed debt against materialized inflation rates, controlling for the presence of fiscal rules, the degree of central bank independence, and a time-fixed effect in an annual sample of all BIS Member Countries issuing inflation-indexed debt.

Inflation-indexed debt therefore does not seem to eliminate inflationary pressures altogether. But how much of this is informed through the link between inflation-indexed debt and the monetary-fiscal policy mix? As alluded to before, and as established rigorously by [Cochrane \(2011\)](#) and [Neumeyer and Nicolini \(2025\)](#), estimating policy rules with empirical data in the hopes of distinguishing between fiscally-led and monetary-led policy mixes is a futile task. We therefore explicitly do not do so, and now mostly present non-causal evidence that suggestively hints at an effect of inflation-indexed debt on measures that can, to a certain extent, be proxies for the possibilities of monetary-led and fiscally-led policy mixes.

A natural proxy evaluating the monetary side is the degree of central bank independence, here measured through the Central Bank Independence Index of [Romelli \(2024\)](#). Intuitively, the index captures the degree to which a central bank is not constrained through other factors (e.g., partisan considerations) in its decision-making. A less independent central bank might hint at a reduced probability of a monetary-led policy mix. Figure C.7 relates this independence index to the share of inflation-indexed debt present at a given country and at a given point in time. In this logic, we therefore follow [Banerjee et al. \(2022\)](#).

While again at best correlational, the evidence is again striking: this time, we observe a clearly negative correlational relationship between the share of inflation-indexed debt in the overall debt stock and the measured degree of central bank independence, in particular across countries since the results are less pronounced when focussing on within-country variation. While not making any causal statements, the observed data hints at a diminished probability of monetary-led policy mixes occurring when large swaths of the government debt stock are indexed to gross inflation.

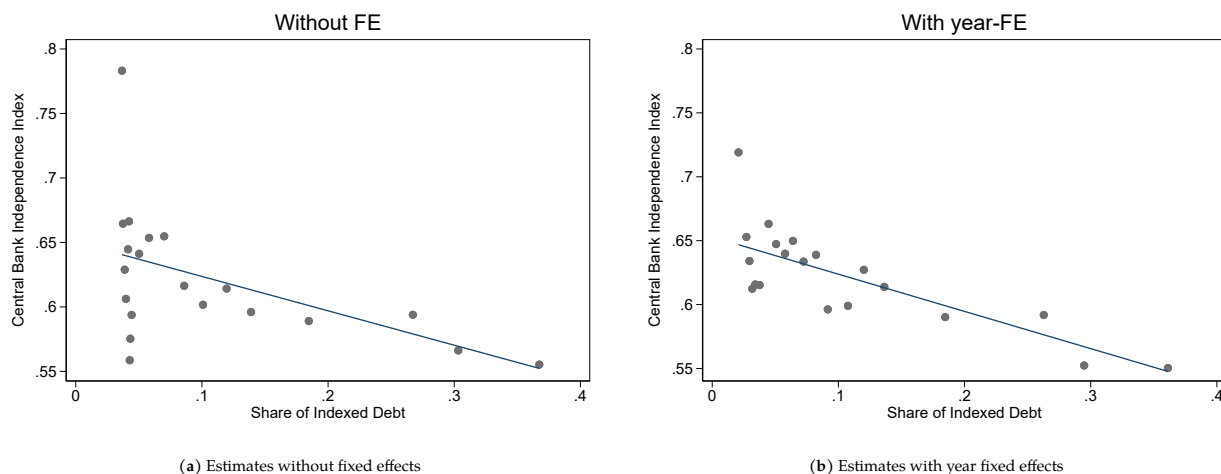


Figure C.7: Binscatter of the relationship between the share of inflation-indexed debt and the measured degree of central bank independence following [Romelli \(2024\)](#).

The following table C.1 provides the numbers corresponding to figure C.7, providing evidence on the correlation between measured central bank independence and the share of inflation-indexed

debt in the overall government debt stock.¹⁶

Dep. var.	Central Bank Independence Index (Romelli, 2024)							
CB board index	0.687*** (0.0194)	0.680*** (0.0201)	0.505*** (0.0152)	0.483*** (0.0169)	0.686*** (0.0192)	0.615*** (0.0187)	0.500*** (0.0149)	0.485*** (0.0164)
Fiscal Rule Intensity	0.0102*** (0.00361)	0.0130*** (0.00397)	0.0118*** (0.00183)	0.00798*** (0.00183)	0.0114*** (0.00370)	0.0447*** (0.00445)	0.00994*** (0.00191)	0.00621*** (0.00181)
Indexed debt share	-0.182*** (0.0413)	-0.198*** (0.0424)	0.258*** (0.0336)	0.105*** (0.0287)	-0.313*** (0.0490)	0.410*** (0.0763)	0.253*** (0.0414)	0.0666** (0.0322)
FisRules × IndexDebt			-0.0893*** (0.0153)			-0.490*** (0.0411)	-0.0955*** (0.0161)	
Inflation					-0.00206* (0.00113)	-0.00125 (0.00108)	-0.00155*** (0.000376)	-0.00222*** (0.000377)
IndexDebt × Inflation					0.0169*** (0.00387)	0.000288 (0.00383)	0.00147 (0.00132)	0.00382*** (0.00123)
Constant	0.251*** (0.0115)	0.252*** (0.0120)	0.326*** (0.00885)	0.344*** (0.0112)	0.260*** (0.0121)	0.253*** (0.0112)	0.337*** (0.00927)	0.365*** (0.0115)
Obs.	605	605	605	605	605	605	605	605
R ²	0.707	0.713	0.978	0.980	0.717	0.778	0.979	0.981
F	482.5	451.3	296.8	41.32	304.1	317.9	210.2	42.46
R ² _{adj}	0.705	0.697	0.977	0.978	0.715	0.765	0.978	0.979
RMSE	0.106	0.107	0.030	0.029	0.104	0.095	0.029	0.028
Year-FE	No	Yes	No	Yes	No	Yes	No	Yes
Country-FE	No	No	Yes	Yes	No	No	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table C.1: Regressions on possible predictors of central bank independence following Romelli (2024)

While establishing a proxy for the monetary side of the policy mix has already been ambitious, guidance for the fiscal side is even more difficult to come by due to the clearly simultaneous nature of fiscal policy setting with overall macroeconomic dynamics (Ramey and Zubairy, 2018). To nonetheless get a semblance of the idea, we exploit data from Davoodi et al. (2022) on the implementation of *fiscal rules* around the globe. While the nature of such rules and their implementation are relatively arbitrary, we zoom into *periods of rule suspension*; that is, periods in which previously established fiscal rules were suspended. In doing so, we focus on the question whether any rule had been suspended, as well as whether specific fiscal rules (expenditure rules, budget balancing rules, and deficit rules) were suspended. While again an imperfect proxy, we interpret here periods of fiscal rule suspension as periods of governments being willing to run fiscal policies that are normally considered unsustainable, therefore contributing to a possible fiscally-led policy mix.

The results of the logit regression predicting factors influencing the probability of a given fiscal rule suspension (for each of the four previously mentioned sets of rule suspensions) are presented in table C.2. We generally observe a clear positive correlation between the share of inflation-indexed debt being present and the probability of rule suspension, except for periods of deficit rule suspension.¹⁷ Taking our most comprehensive estimation in column five, we can observe that each percentage point increase in the share of inflation-indexed debt is correlated with a 0.634% increase

¹⁶In appendix C.3, we provide some further robustness exercises.

¹⁷The cross-country variation on the suspension of debt rules is quite low, however, impeding the statistical power of the estimation.

in the probability of the suspension of any fiscal rule that had previously been put in place.

Dep. var.	AnyR	ER	BR	DR	AnyR	ER	BR	DR
CB independence	-2.2267 (1.8226)	-3.5126 (3.9415)	-2.9157 (2.0313)	0.4831 (4.9218)	0.6227 (2.1084)	13.5366 (10.9178)	0.0185 (1.9724)	0.9860 (4.8700)
CB board index	1.6759 (1.5110)	4.8314 (3.7678)	1.7574 (1.7029)	-0.2362 (3.0294)	-1.2981 (1.4926)	-6.5182 (6.1250)	-1.5719 (1.4332)	-1.2533 (2.7346)
Indexed debt share	3.0572*** (0.6578)	4.7854*** (1.0479)	3.1249*** (0.6751)	0.3678 (0.9102)	6.3852*** (1.1965)	13.6669*** (4.0720)	6.4125*** (1.2509)	2.0403 (1.3204)
Inflation	-0.0162 (0.0154)	-0.0255 (0.0219)	-0.0767* (0.0379)	-0.1554* (0.0674)	-0.0204 (0.0775)	-0.1393** (0.0432)	-0.1335** (0.0458)	-0.2160* (0.0958)
Constant	-3.1932*** (0.5008)	-5.6121*** (0.5327)	-2.7544*** (0.4616)	-4.1307** (1.5229)	-1.1790 (0.9511)	-8.8706* (4.1165)	-0.4267 (0.8817)	-1.0932 (1.8025)
Obs.	652	652	652	652	285	261	285	95
ll	-103.8148	-51.7716	-91.2659	-42.2145	-59.0467	-18.4910	-56.0453	-24.6104
χ^2	27.9986	66.0330	27.5217	16.0084	70.0490	69.0798	70.0949	8.8445
p	0.0000	0.0000	0.0000	0.0030	0.0000	0.0000	0.0000	0.2640
R^2	0.1014	0.2750	0.1122	0.0218	0.3550	0.6779	0.3194	0.1034
Year-FE	No	No	No	No	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.2: Rule Suspension Regression. The outcome variables are defined as: *AnyR* = “Any Fiscal Rule Suspended”, *ER* = “Expenditure Rule Suspended”, *BR* = “Budget Balancing Rule Suspended”, *DR* = “Deficit Rule Suspended”.

To visualize the results of table C.2, figure C.8 predicts the probability of the suspension of any fiscal rule as a function of the share of inflation-indexed debt. To maximize insights, figure C.8 performs this analysis at both higher inflation levels of 5% annualized CPI inflation, as well as for lower levels of 1% annualized CPI inflation.

Irrespective of the level of inflation, figure C.8 shows a general clear positive link between the share of inflation-indexed debt and the propensity of the suspension of any fiscal rule. That being said, the effect seems to be quite limited for most advanced economies, which boast inflation-indexed debt shares of below 30%.¹⁸ There is, however, a significant difference between low-inflationary and high-inflationary environments in the propensities of a rule suspension, with the probability being higher in *low* inflation environments. While this might seem surprising, it can easily be rationalized as high-inflation environments are usually characterized by a lack of *surprise* inflationary pressure, as higher inflation levels usually imply that corresponding market expectations (and therefore market prices of inflation-indexed debt) reflect expectations of pronounced inflationary episodes. But inflation-indexed debt becomes especially costly during surprise inflation episodes, which must arise from an initial point of relative price stability.

Even though the non-causal evidence presented here is useful to set the ideas, we now intend to move on to the degree possible and provide a semblance of causal evidence on the role that inflation-indexed debt might play in shaping inflationary dynamics across policy regimes. While identifying policy regimes is generally hardly possible (meaning that evaluating the effect of in-

¹⁸Note, however that up to 80% of debt being indexed to factors other than the price level is possible. For instance, many emerging market economies issue large amounts of debt denominated to foreign currencies, which work tantamount to inflation-indexed debt in simple models of the Fiscal Theory of the Price Level (Cochrane, 2023).

dexed debt on the propensity of either policy regime in a causal sense is prohibitively difficult), we can attempt to find evidence on the inflationary consequence of inflation-indexed debt in either policy regime, conditional on supplying the prevalence of each regime exogenously.

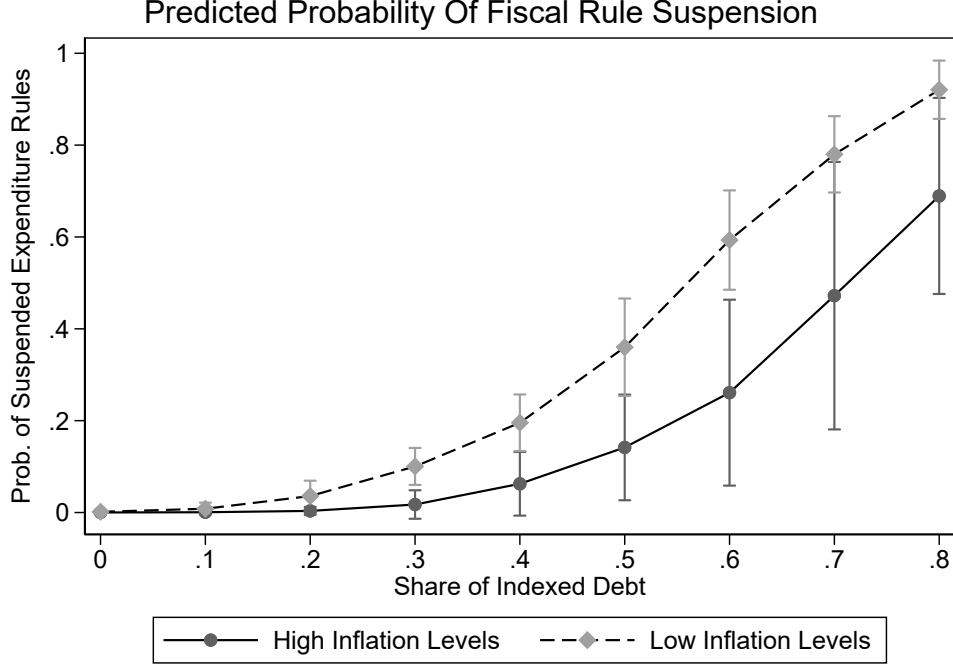


Figure C.8: The relationship between the share of inflation-indexed debt and the predicted probability of a suspension of fiscal rules in high- and low-inflationary environments, following the estimation in table C.2.

C.2 Causal evidence on the link between fiscally-led policy mixes and materialized inflation in the light of high shares of indexed debt

To round up our empirical exercise, we here present direct evidence on the effect that inflation-indexed debt can have on inflation, making use of the series of narratively identified tax shocks in the US provided by [Mierzwa \(2024\)](#).

We leverage his time series of exogenous fiscal policy surprises, and combine it with a corresponding series of inflation-indexed debt, taking the share of inflation-indexed debt in the overall sovereign debt portfolio as our main indicator for the intensity of the prevalence of inflation-indexed debt. Equipped with these time series, we estimate the following local projection ([Jordà, 2005](#)) to measure the dynamic impact of inflation-indexed debt on changes in the rate of inflation:

$$\log P_{t+h} - \log P_{t-1} = \alpha_h + \beta_h \omega_{t-1} \epsilon_t^F + \delta_{1h} \omega_{t-1} + \delta_{2h} \epsilon_t^F + \Gamma_h Z_{t-1} + e_{t+h}, \quad (\text{C.1})$$

where $h \geq 0$ indexes the forecast horizon considered and Z_{t-1} is a vector of control variables specified below. Of particular interest to us is the coefficient β_h , which captures the cross-effect of

the identified fiscal shock ε_t^F and the past level of the share of inflation-indexed debt ω_{t-1} .¹⁹ We estimate this equation separately for two different periods, creating effectively a *state-dependent local projection*, as described by [Jordà and Taylor \(2025\)](#). We estimate the specification (C.1) separately for periods of *fiscally-led policy mixes* and *monetary-led policy mixes*, as identified by [Chen et al. \(2022\)](#) for the United States.²⁰ We label the regimes here ‘active fiscal policy’ and ‘passive fiscal policy’ in line with [Chen et al. \(2022\)](#).

Figure C.9 summarizes this exercise, with the left-hand panel showcasing the fiscally-led policy mix and the right-hand panel showing the monetary-led policy mix. Under the monetary-led policy mix, the response of the price level on the interaction between the share of inflation-indexed debt and the identified fiscal policy shock is relatively imprecisely estimated, but - if anything - there is at least a temporary negative effect on the price level when more debt is indexed. In the case of a fiscally-led policy mix, however, medium-term price pressures arising from the identified fiscal policy disturbance are significantly increasing in the share of inflation-indexed debt. In the period 4-5 years after the initial shock, a 1% deficit-to-GDP shock accompanied by a corresponding level difference of 1% in the share of inflation-indexed debt boosts observed changes to the price level by up to 1.5% relative to a case where the share of inflation-indexed debt is lower. This is an economically meaningful effect that would remain hidden without consideration of inflation-indexed debt for inflationary dynamics.

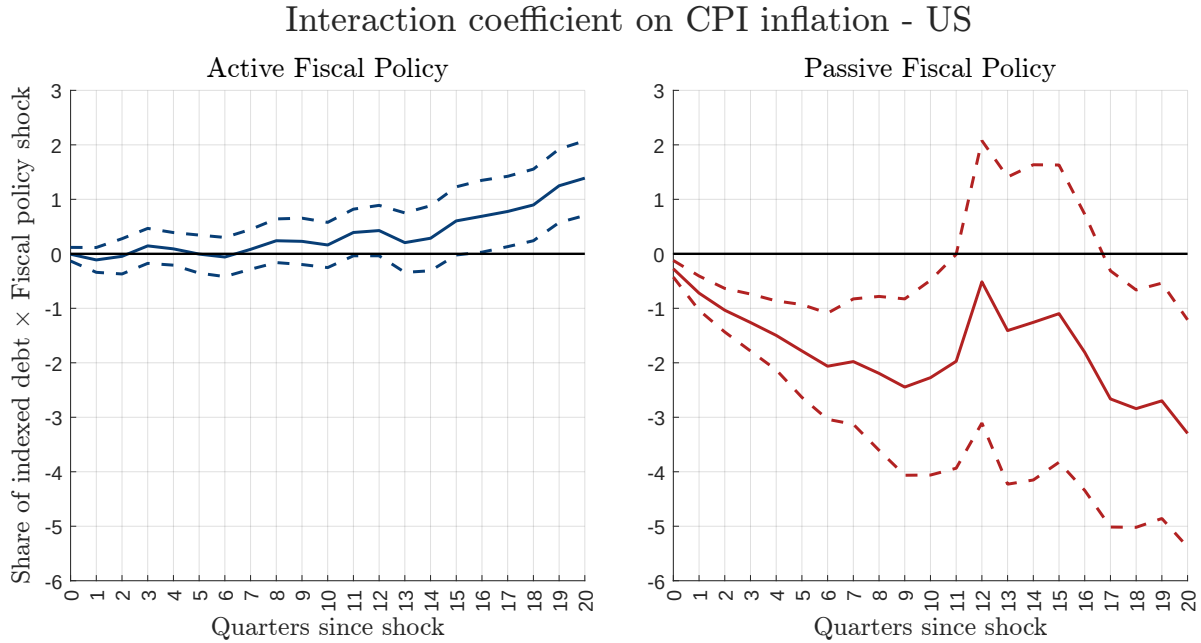


Figure C.9: IRF implied by the local projection (C.1) through the coefficients β_{it} . The control vector Z contains the first four lags of the real GDP growth rate, the short-run Fed Funds rate, the change in the weighted real exchange rate, a same-period recession indicator, year-fixed effects, and the first lag of the price level difference. Standard errors are robust to heteroscedasticity and autocorrelation (Newey-West correction). Confidence intervals are provided at the 90% level. Sample length: 1970 Q1 - 2019 Q2.

¹⁹We use last period’s level of inflation-indexed debt to avoid running into a simultaneity bias.

²⁰While the results in their paper only extend for the period until the Great Financial Crisis, we utilize their characterization to further pin down the prevalent policy mix until 2020, inclusive. Our results are qualitatively robust to characterizing the prevalent fiscal-monetary regime as [Bianchi and Melosi \(2022\)](#) do.

C.3 Additional empirical results

First, we replicate the results from the beginning of this appendix without controlling for the central bank board index of [Romelli \(2024\)](#), which is closely related to the central bank independence index itself.

Dep. var.	Central Bank Independence Index (Romelli, 2024)							
Fiscal Rule Intensity	0.0456*** (0.0061)	0.0511*** (0.0066)	0.0129*** (0.0031)	0.00170 (0.0029)	0.0502*** (0.0063)	0.1039*** (0.0069)	0.0102** (0.0033)	-0.00030 (0.0029)
Indexed debt share	-0.1845* (0.0726)	-0.2348** (0.0734)	0.2445*** (0.0574)	-0.0057 (0.0449)	-0.3400*** (0.0867)	0.9565*** (0.1269)	0.2276** (0.0712)	-0.0638 (0.0513)
FisRules × IndexDebt			-0.1095*** (0.0260)			-0.8851*** (0.0670)	-0.1155*** (0.0277)	
Inflation					0.0014 (0.0020)	0.0030 (0.0018)	-0.0022*** (0.0006)	-0.0022*** (0.0006)
IndexDebt × Inflation					0.0135* (0.0068)	-0.0176** (0.0065)	0.0025 (0.0023)	0.0051** (0.0020)
Constant	0.5720*** (0.0124)	0.5677*** (0.0130)	0.5991*** (0.0056)	0.5554*** (0.0133)	0.5642*** (0.0152)	0.4938*** (0.0144)	0.6127*** (0.0074)	0.5777*** (0.0145)
Obs.	605	605	605	605	605	605	605	605
R ²	0.0938	0.1393	0.9355	0.9494	0.1134	0.3539	0.9373	0.9506
F	31.1719	35.0722	9.2438	6.1690	19.1892	56.5439	8.9450	6.3173
R ² _{adj}	0.0908	0.0943	0.9326	0.9445	0.1075	0.3166	0.9342	0.9457
RMSE	0.1856	0.1852	0.0505	0.0458	0.1839	0.1609	0.0499	0.0454
Year-FE	No	Yes	No	Yes	No	Yes	No	Yes
Country-FE	No	No	Yes	Yes	No	No	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.3: Regressions on possible predictors of central bank independence (no CB board control), following [Romelli \(2024\)](#).

Tables [C.4](#) and [C.5](#) show, respectively, the results when ‘we do not control for the composition of central bank board members and when we include also countries that never issued any inflation-linked government bonds in the sample period.

Dep. var.	AnyR	ER	BR	DR	AnyR	ER	BR	DR
CB independence	-0.9230 (0.9653)	-0.3724 (1.9799)	-1.5230 (0.9818)	0.2394 (1.9364)	-0.4363 (1.2777)	6.8147** (2.3326)	-1.2566 (1.2063)	-0.2508 (2.3983)
Indexed debt share	3.0495*** (0.6649)	4.5955*** (0.9761)	3.1201*** (0.6850)	0.3687 (0.8924)	6.2171*** (1.1265)	12.0975*** (1.5091)	6.1895*** (1.2104)	1.9829 (1.3227)
Inflation	-0.0136 (0.0144)	-0.0173 (0.0181)	-0.0721* (0.0353)	-0.1561* (0.0651)	-0.0167 (0.0812)	-0.1236*** (0.0255)	-0.1319** (0.0443)	-0.2138* (0.0975)
Constant	-3.0944*** (0.5391)	-4.7794*** (0.8917)	-2.6751*** (0.5229)	-4.1030*** (1.1886)	-1.2264 (0.9449)	-8.0511*** (1.7184)	-0.4638 (0.8612)	-1.0064 (1.5822)
Obs.	652	652	652	652	285	261	285	95
ll	-104.2834	-53.1646	-91.7183	-42.2181	-59.2459	-19.6852	-56.3336	-24.7103
χ^2	24.9541	80.2726	24.3913	12.9324	72.2432	90.1966	71.7882	7.4952
p	0.0000	0.0000	0.0000	0.0048	0.0000	0.0000	0.0000	0.2775
R^2	0.0973	0.2555	0.1078	0.0217	0.3528	0.6571	0.3159	0.0998
Year-FE	No	No	No	No	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.4: Rule Suspension Regression. The outcome variables are defined as: *AnyR* = “Any Fiscal Rule Suspended”, *ER* = “Expenditure Rule Suspended”, *BR* = “Budget Balancing Rule Suspended”, *DR* = “Deficit Rule Suspended”.

Dep. var.	Central Bank Independence Index (Romelli, 2024)							
CB board index	0.5595*** (0.0085)	0.5172*** (0.0083)	0.6292*** (0.0075)	0.5358*** (0.0077)	0.5654*** (0.0086)	0.5114*** (0.0084)	0.6235*** (0.0076)	0.5255*** (0.0078)
Fiscal Rule Intensity	0.0164*** (0.0015)	0.0096*** (0.0015)	0.0198*** (0.0012)	0.0095*** (0.0012)	0.0139*** (0.0015)	0.0091*** (0.0016)	0.0202*** (0.0012)	0.0092*** (0.0012)
Indexed debt share	-0.1972*** (0.0389)	-0.2759*** (0.0373)	0.4347*** (0.0419)	0.1838*** (0.0292)	-0.3216*** (0.0470)	0.0779 (0.0745)	0.4583*** (0.0523)	0.1565*** (0.0336)
FisRules × IndexDebt			-0.0903*** (0.0235)			-0.3081*** (0.0380)	-0.0991*** (0.0256)	
Inflation					0.0000 (0.0000)	0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)
IndexDebt × Inflation					0.0123*** (0.0029)	0.0013 (0.0031)	-0.0013 (0.0019)	0.0017 (0.0016)
Constant	0.3144*** (0.0044)	0.3415*** (0.0043)	0.2725*** (0.0037)	0.2826*** (0.0065)	0.3184*** (0.0044)	0.3505*** (0.0044)	0.2798*** (0.0038)	0.2879*** (0.0068)
Obs.	5568	5568	5568	5568	5282	5282	5282	5282
R^2	0.5092	0.5558	0.8560	0.8769	0.5166	0.5736	0.8549	0.8769
F	1924.2808	1554.2882	2458.8637	300.8337	1127.4948	768.2458	1580.1035	279.3330
R^2_{adj}	0.5089	0.5525	0.8518	0.8724	0.5161	0.5700	0.8505	0.8723
RMSE	0.1167	0.1114	0.0641	0.0595	0.1160	0.1094	0.0645	0.0596
Year-FE	No	Yes	No	Yes	No	Yes	No	Yes
Country-FE	No	No	Yes	Yes	No	No	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.5: Regressions on possible predictors of central bank independence following Romelli (2024). Zero-imputation applied to missing values of IndexedDebtShare, i.e., we use all countries here, including those who never issued any indexed debt.

Rule suspension regressions against lagged levels of indexed debt

Dep. var.	AnyR	ER	BR	DR	AnyR	ER	BR	DR
CB independence	-0.9616 (0.9171)	-0.2181 (1.9145)	-1.5826 (0.9429)	0.1545 (1.9678)	-0.4310 (1.2830)	7.7258*** (2.2862)	-1.2557 (1.2018)	-0.2000 (2.4362)
<i>L</i> . Indexed debt share	2.9801*** (0.6474)	4.5868*** (0.9392)	3.1124*** (0.6645)	0.5533 (0.8735)	6.4393*** (1.1373)	12.5273*** (1.6212)	6.0911*** (1.1368)	2.0002 (1.1816)
<i>L</i> . Inflation	-0.0095 (0.0198)	-0.0130 (0.0253)	-0.0756* (0.0356)	-0.1814*** (0.0541)	0.0346 (0.1189)	-0.0858** (0.0274)	-0.1113** (0.0409)	-0.1331** (0.0507)
Constant	-3.0050*** (0.5224)	-4.8858*** (0.8759)	-2.5595*** (0.5040)	-3.9959*** (1.1948)	-1.1878 (0.9295)	-9.2778*** (1.7401)	-0.6361 (0.9002)	-1.6762 (1.5650)
Obs.	632	632	632	632	286	262	286	96
<i>ll</i>	-106.3518	-52.3364	-93.5731	-41.7471	-58.9689	-18.6168	-57.3783	-25.2193
χ^2	24.9385	81.4065	24.3770	24.1283	71.4016	75.5276	79.6409	11.7768
p	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0671
R^2	0.0963	0.2622	0.1109	0.0270	0.3717	0.6760	0.3234	0.0841
Year-FE	No	No	No	No	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.6: Rule Suspension Regression with lagged regressors. Specification includes *L*.Indexed debt share and *L*.Inflation. Outcomes: *AnyR* (Any Rule Suspended), *ER* (Expenditure Rule), *BR* (Budget-Balance Rule), *DR* (Deficit Rule). Columns 1–4 exclude year fixed effects; columns 5–8 include year fixed effects (coefficients not shown).

Dep. var.	AnyR	ER	BR	DR	AnyR	ER	BR	DR
CB independence	-0.0808 (1.1800)	3.7843* (1.8603)	-0.5463 (1.2741)	0.2099 (2.1038)	-0.0274 (1.9263)	8.6885** (2.7690)	-0.7570 (1.8040)	0.0139 (3.3163)
<i>L</i> . Indexed debt share	1.7687* (0.7415)	3.1557*** (0.9130)	1.8284* (0.8265)	0.6844 (0.9016)	3.2555** (1.1305)	7.4810*** (1.7710)	2.8992** (1.1188)	2.3885 (1.7064)
<i>L</i> . Inflation	-0.0045 (0.0393)	-0.0055 (0.0605)	-0.0376 (0.0283)	-0.1959** (0.0723)	0.0430 (0.0534)	0.0052 (0.0298)	-0.0569* (0.0277)	-0.1041 (0.0548)
Lagged Rule	4.2898*** (0.5175)	5.2069*** (0.8183)	4.5895*** (0.5701)	4.9449*** (0.8629)	0.0000 (.)	0.0000 (.)	0.0000 (.)	0.0000 (.)
Constant	-4.1893*** (0.7992)	-8.3652*** (1.2691)	-3.9760*** (0.8350)	-4.7069*** (1.3290)	-3.2621 (1.6873)	-9.3682*** (2.0013)	-2.5648 (1.5199)	-1.8548 (1.9630)
Obs.	632	632	632	632	286	48	286	48
<i>ll</i>	-70.8248	-28.8303	-59.8161	-29.0105	-28.9850	-7.1589	-28.0778	-12.5956
χ^2	81.0217	66.7653	75.5655	47.9633	.	29.0637	.	5.1460
p	0.0000	0.0000	0.0000	0.0000	.	0.0000	.	0.2726
R^2	0.3982	0.5936	0.4317	0.3238	0.6912	0.4800	0.6689	0.0852
Year-FE	No	No	No	No	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. “LDV” denotes the lag of the column’s dependent variable; dots (.) indicate omission.

Table C.7: Rule Suspension Regression with lagged dependent variable (LDV), lagged indexed debt share, and lagged inflation (no interaction terms). Outcomes: *AnyR* (Any Rule Suspended), *ER* (Expenditure Rule), *BR* (Budget-Balance Rule), *DR* (Deficit Rule). Columns 1–4 exclude year fixed effects; columns 5–8 include year fixed effects (coefficients not shown).